Do industries matter in explaining stock returns and asset-pricing anomalies?☆

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Abstract

Industry returns can not be explained fully by well-known asset pricing models. This study reveals that common factors extracted from industry returns carry significant risk premiums that go beyond the explanatory power of size, book-to-market (BM) ratios, and momentum. In particular, this study shows that (1) the small-firm effect is significant only for firms whose market capitalization is below their industry average; (2) the BM effect is an intra-industry phenomenon; (3) a one-year momentum effect is significant only for firms whose past one-year return is smaller than the industry average and limited to non-January months; and (4) there is seasonality in all effects that cannot be explained by risk-based asset-pricing models. Neither rational nor behavioral theories alone can explain industry returns, and it is perhaps too hasty to attribute asset pricing anomalies to a single driving force.

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1. Introduction

Despite its popularity in practice, industry analysis has received limited academic attention in finance. Microeconomics holds that the market supply of a product is determined by a group of firms that produce homogenous products (i.e., industry). But in financial economics, the supply (or demand) of an asset is infinitely elastic, because all assets are perfect substitutes. Popular models, whether rational or behavioral, simply grant no role to industries.[1] Nevertheless, researchers commonly control for the "industry effect" in empirical studies, without any theoretical foundation for doing so (e.g., Kahle and Walking, 1996).[2]

Theoretically, an industry refers to a group of firms producing homogenous products or close substitutes; practically, a firm rarely produces just one product. Broad industry classifications, such as standard industrial classification (SIC) codes, thus have been used widely to identify homogeneous groups of firms that engage in practice in "close" businesses. These classification schemes generally reflect broad attributes, such that firms of the same industry may be competitive when they produce close substitutes but cooperative when their products are complements. Firms, even in the same industry, therefore may respond differently to information, whether it is market-wide, industry-specific, or firm-specific in nature. Because the product mixes or even the business units of a firm could span a wide range, both vertically and horizontally, it is difficult to foresee how firms might respond to relevant information. In this sense, industry classifications such as SIC codes are far from satisfactory.[3]

The discrepancy in the definitions of industry casts doubt on the applicability of the

[1]For example, consider the one-factor Sharpe-Lintner-Black capital asset pricing model (CAPM), the macroeconomic-based model of Chen, Roll, and Ross (1986), the three-factor model by Fama and French (1993), or the characteristic-based model advocated by Daniel and Titman (1997), among others. Although theoretical studies have examined the impact of industry structure on capital structure or financial structure (e.g., MacKay and Phillips, 2005; Miao, 2005), standard asset-pricing theories suggest that neither technical nor fundamental analysis is important.

[2]Kahle and Walkling (1996) identify 81 articles published in the five top-tier finance journals during a four-year sample period (1992-1995) that use industry classifications.

[3]Bhojraj, Lee, and Oler (2003) and Chan, Lakonishok, and Swaminathan (2007) provide comparisons of alternative industry classification schemes.

micro-based industry analysis. Does a practical industry classification system actually provide any information about stock returns? Theoretical evidence indicates that industry structure affects capital structure, but does it also affect asset prices? If industries really matter in asset pricing, how and why are they related to asset prices? In particular, does industry-related information help explain asset pricing anomalies such as size, book-to-market (BM) ratios, and momentum? We explore such questions in this article.

Our motivation stems from recent research that has identified industry-related patterns that standard asset pricing models cannot explain effectively. For example, Fama and French (1997) find that neither the CAPM nor their three-factor model provides precise estimates for the industry cost of equity. Lewellen, Nagel, and Shanken (2010; hereafter LNS) show that several risk-based asset pricing models are rejected because they fail to explain the cross-section of returns on industry portfolios. Hou and Robinson (2006) reveal that firms in concentrated industries earn lower returns, even after they control for size, BM, and momentum. Moskowitz and Grinblatt (1999) also show that individual stock momentum is largely driven by industry momentum and that stocks within an industry tend to be more highly correlated than stocks across industries.[4] Finally, according to Chan, Lakonishok, and Swaminathan (2007), higher return comovement is more pronounced for large-cap stocks that belong to the same industry classification compared with that for small-cap stocks of the same industry.

Such higher within-industry return comovements might be driven by rational or behavioral forces. Rationally, firms of the same industry exhibit higher return comovements because they share more common fundamentals. In this case, large firms lead small firms of the same industry because the former respond to information more quickly. Hou (2007) confirms that the lead-lag effect is predominantly an intra-industry phenomenon and also drives the industry momentum anomaly. The higher within-industry return comovements can also behaviorally driven if industries were treated as styles by investors (see Barberis

[4]In contrast, Grundy and Martin (2001), Lewellen (2002) and Wang and Wu (2011) assert that industry effects do not explain momentum.

and Shleifer, 2003; Barberis, Shleifer, and Wurgler, 2005; Kumar and Lee, 2006). In this case, the return comovements reflect non-fundamental forces, such as investor sentiment, that induce negative lead-lag relations among securities.

How do higher within-industry comovements relate to asset pricing? Intuitively, return comovements within an industry imply potentially nondiversifiable risk. Asset pricing models such as the CAPM or the arbitrage pricing theory (APT, Ross, 1976) suggest that assets are correlated through their relations to the market portfolio or common factors. If existing pricing factors fail to capture excess within-industry comovements, additional common factors might be needed, and the industry-related comovements must represent a non-negligible proportion of the variation in stock returns. In contrast, if the excess within-industry comovements are behavioral, no systematic pricing would be associated with the industry-related patterns.

Although existing empirical evidence suggests that industry plays a role in stock returns, it is not clear if the industry-related patterns are consistent with standard asset pricing theories. The first objective of this study therefore is to explore the role of industry in explaining the cross-section of stock returns from rational viewpoints.

Specifically, we examine the role of industry in an APT framework. Motivated by Connor and Korajczyk (1988) who propose the use of asymptotic principal components to extract common factors from individual stock returns, we use principal components analysis to extract various factors from industry portfolios, and examine whether they bear significant factor risk premiums. We find that two industry-based risk factors, constructed on the last two of the five principal components, bear significant risk premiums in explaining the cross-section of stock returns. However, based on the Fama-MacBeth (1973) cross-sectional regression, we find that the industry-based factors do not subsume the explanatory ability of size, BM, or momentum.

In a stochastic discount factor (SDF) setting, we also examine the validity of various asset-pricing models based on Hansen's test of overidentifying restrictions, along with the Gibbons-Ross-Shanken (1989; hereafter GRS) F-test to determine if there are significant deviations from the pricing relation implied by the pricing models. The results indicate that all models, including the Sharpe-Lintner-Black CAPM, the Fama-French three-factor model, Carhart's (1997) four-factor model, and two industry-related factor models, fail to fully explain returns on an extended sample composed of the 25 size-BM portfolios and 48 industry portfolios.

Since industries as rational factors cannot explain the asset pricing anomalies, we turn to potential behavioral explanations and explore how industries interact with firm characteristics on size, BM, and past returns. Motivated by the empirical evidence that the momentum effect is an intra-industry phenomenon (Moskowitz and Grinblatt, 1999; Hou, 2007), we examine if the premiums on size, BM, and momentum are the same for firms within and across industries. Motivated as well by recent behavioral evidence regarding risk attitudes toward gains and losses (e.g., Kahneman and Tversky, 1979), we also examine if the premiums on size, BM, and momentum exhibit asymmetric patterns for firms whose characteristics rank them above or below their industry averages. For example, Fiegenbaum and Thomas (1988) and Fiegenbaum (1990) document a negative (positive) association between risk and return for firms whose accounting returns fall below (above) the industry median. To the extent that size, BM, and past returns reflect a firm's future prospects, premiums on these firm characteristics may exhibit asymmetric patterns over some industry reference points.

The empirical evidence reveals interesting patterns that appear inconsistent with risk-based theories. First, the small-firm premium is significant only for firms whose market capitalization falls below their industry average. Thus, the size effect is essentially a "below-industry" phenomenon. Second, the BM premium is significant only for stocks within an industry, not for stocks across industries, which means the value effect is an intra-industry phenomenon. Third, the one-year momentum effect is significant only for firms across industries, yet this across-industry effect disappears when we adjust the returns for risk. What remains is a below-industry momentum effect for 11 months of the year but reveals a strong reversal in January. In turn, we identify a January-based seasonality in size and BM effects that we cannot explain using popular risk-based asset-pricing models.

Summarizing, we identify several unique industry-related patterns that appear to be new in the literature. The empirical findings indicate that industry returns reflect both significant rational and behavioral components, but neither rational nor behavioral theories alone can fully explain industry returns. The results indicate that industries play a dual role in explaining stock returns that deserves further exploration.

The remainder of this article proceeds as follows: Section 2 describes the sample. Section 3 reports the correlations for stocks within and across industries over various return intervals. In Section 4, we present the empirical results regarding the validity of various asset pricing models for explaining returns on industry portfolios. In Section 5, we present the empirical results on behavioral patterns, then we combine our analysis to consider the interaction of rational and behavioral roles in industry returns. After outlining some robustness tests in Section 7, we conclude with a summary of our findings.

2. Data

The data used in this study are ordinary common equities of all firms listed on the NYSE, AMEX, and NASDAQ return files from the Center for Research in Security Prices (CRSP) from July 1963 (1973 for NASDAQ firms) to December 2006. The accounting data come from the COMPUSTAT database. To mitigate the survivorship bias, we require that firm data must be available on COMPUSTAT for at least two years (Fama and French, 1993).

Firm size and BM are calculated as in Fama and French (1992, 1993). Specifically, a firm's market equity in June of year t indicates its size for July of year t to June of year t+1. A firm's book-to-market equity for July of year t to June of year t+1 is the book value of fiscal year t-1, divided by market equity (ME, equal to stock price times shares outstanding) at the end of calendar year t-1. The book value is the stockholder's equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. Firms with negative book values are excluded. As is common practice, both size and BM are transformed by taking a natural logarithm.

We obtain the industrial classifications from Kenneth French's Web site.[5] Each month, we assign all firms listed on the NYSE, AMEX, and NASDAQ into one of the 48 industries based on the four-digit SIC codes obtained from COMPUSTAT.[6] Value-weighted monthly returns for industry portfolios are also downloaded from French's website.

Table 1 presents summary statistics for each of the 48 industry portfolios. The average number of firms in an industry varies from 5.9 to 388.7, suggesting that investing capital in a single industry might not be well diversified. The average monthly returns for industry portfolios range from 0.73% to 1.62%, with a grand average of 1.11% for all industries. As a preliminary analysis, we also calculate the abnormal return for each portfolio based on the Fama-French three-factor model. Specifically, the abnormal return of an industry portfolio is the intercept of a time-series regression of its monthly excess returns on the Fama-French three factors over the full sample period. The last row of Table 1 indicates that the average monthly abnormal return across all industries is -0.06%. According to the exact GRS test, we reject the null hypothesis that the abnormal returns are zero for all industries; the F-statistic is 2.29 with a p-value smaller than 0.01. Thus, the preliminary result indicates that returns on industry portfolios cannot be well explained by the popular Fama-French three-factor model. However, the result may not be robust, because the regressions assume constant risks over 1963-2006, a sample period that spans more than 40 years.

[Insert Table 1 about here]

3. Correlation structure within and across industries over various return intervals

If industries matter, in the sense that stocks of the same industry share more commonalities than those of different industries, we expect correlations to differ for stocks within and across industries. We establish the following predictions:

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[5] See http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/.

[6] Kahle and Walkling (1996) point out large differences between CRSP and COMPUSTAT databases in terms of SIC codes. More than 36 percent of the classifications disagree at the two-digit level, and nearly 80 percent disagree at the four-digit level. Following French's definition, we use COMPUSTAT SIC codes, complemented with the CRSP SIC codes when the COMPUSTAT codes are not available.

1. If the SIC classification scheme is "correct," differences in correlations between within-industry stocks and across-industry stocks increase with finer industry classifications.
2. If the commonalities shared by stocks of the same industry come from their fundamentals, higher correlations occur among stocks of the same industry than among those of different industries. The correlation structure will be relatively stable for different return intervals.
3. If the commonalities shared by stocks of the same industry are driven by investor sentiment, within-industry correlations will be higher than across-industry correlations. However, the within-industry correlations should decline as the return interval increases, that is, as the nonrational correlations eventually are corrected.

Let $R_i(t, t{+}k)$ be the cumulative k-period returns from time t to $t{+}k$. The k-period correlation between two stocks, i and j, is calculated using overlapping observations:

$$\hat{\rho}_{ij}(k) = \frac{\hat{\sigma}_{ij}(k)}{\hat{\sigma}_i(k)\hat{\sigma}_j(k)},$$

where

$$\hat{\sigma}_{ij}(k) = \frac{1}{T-k-1}\sum_{s=1}^{T-k}(R_i(s, s+k) - \overline{R_i(k)})(R_j(s, s+k) - \overline{R_j(k)}),$$

$$\hat{\sigma}_i(k) = \left(\frac{1}{T-k-1}\sum_{s=1}^{T-k}(R_i(s, s+k) - \overline{R_i(k)})(R_i(s, s+k) - \overline{R_i(k)})\right)^{\frac{1}{2}},$$

and $\overline{R_i(k)}$ refers to the sample average of the k-month holding returns for stock i.

We calculate the intra- and inter-industry correlations with different classifications of industries that we derive through different groupings of SIC codes. The number of industries ranges from 5, which is the broad grouping obtained from French's website, to 410, or the four-digit SIC codes. For all sample stocks, we calculate correlations for each pair of stocks with sufficient observations over the full sample period. We then calculate the average within-industry and across-industry correlations, as displayed graphically in Figure 1.[7]

[7]The numerical results are omitted for brevity. They are available upon request.

[Insert Figure 1 about here]

First, for any given return interval, both the within-industry and across-industry correlations increase as the industry classification becomes finer. For example, for a one-month return interval (i.e., $k = 1$), the average within-industry (across-industry) correlation increases from 0.1162 (0.0948) with the five-industry grouping to 0.1589 (0.0989) with the four-digit SIC grouping. In addition, the average within-industry correlation is always greater than the across-industry correlation. The within-industry correlation ranges from 0.1162 to 0.2367, while the across-industry correlation ranges from 0.0948 to 0.1076. Thus, SIC codes do provide significant information.

Second, the average within-industry correlation increases as the return interval lengthens, whereas the across-industry correlation remains relatively stable over different return intervals. For example, for the four-digit SIC grouping, the average within-industry correlation increases from 0.1589 ($k = 1$) to 0.2367 ($k = 12$). The higher within-industry correlation apparently can be attributed to the commonalities in fundamentals shared by stocks of the same industry. The increase in correlation as the return interval lengthens also suggests nondiversifiable risks related to industry returns.

To determine if the higher within-industry correlation is actually nondiversifiable, we reestimate the correlation structure using the residuals from the regression of the Fama-French three-factor model. We first regress the returns of each stock on the Fama-French three-factor model, then use the residuals to reestimate the within- and across-industry correlations. If industries provide additional information beyond the existing factor models, the same pattern should emerge be retained when we replace the return with the residual. The results in Table 2 reveal two interesting findings.

[Insert Table 2 about here]

First, the average within-industry correlation drops significantly, especially for shorter return intervals with closer industry groupings. For the monthly return interval ($k = 1$), the average within-industry correlation is 0.1162, but it drops to 0.0165 for residuals. Second, the across-industry residual correlation is very close to 0, which indicates that the inter-industry correlation is well explained by existing factor models.

Thus, a large proportion of short-term correlation can be explained by existing factor models. However, the residual within-industry correlation is still 0.088 for the classification based on four-digit SIC codes, so our findings suggest that industry portfolios offer significant information beyond the existing factors.

4. Rational perspectives

This section explores the role of industry returns from rational perspectives. Noting the stronger return comovement for stocks of the same industry, we propose that industry portfolios may reflect some missing risk factors. It is of interest to examine whether it is possible to extract additional factors from returns on industry portfolios that have incremental pricing ability over the existing models.

Firm prospects depend on its firm-specific characteristics, the structure of the industry to which it belongs, and the state of the economy. Firms of different industries thus may have different sensitivities to business cycles (i.e., macroeconomic factors), according to products they produce and the different stages of their industry life cycle. Returns on industry portfolios therefore could convey information about the state of the economy. Lamont (2001) and Hong, Torous, and Valkanov (2007) find that some combinations of industry portfolios can forecast several macroeconomic variables, sometimes leading them by up to two months. Moreover, industry returns provide forecasts of various indicators of economic activity, such as industrial production growth.[8] Hou and Robinson (2006) find that the premium for industry concentration exhibits systematic business cycle variation. Barriers to entry in highly concentrated industries may insulate firms from aggregate shocks that

[8] Lamont (2001) uses eight industry-sorted stock portfolios, plus the stock market portfolio and four bond portfolios, as "base assets" to construct economic tracking portfolios that track macroeconomic variables. He finds that base assets help forecast U.S. output, consumption, inflation, labor income, inflation, stock returns, bond returns, and Treasury bill returns. Hong, Torous, and Valkanov (2007) investigate whether the returns of industry portfolios can predict the movements of the aggregate stock market. Retail, services, commercial real estate, metal, and petroleum portfolios lead the stock market by up to two months.

lead to economic distress. Overall, the empirical evidence suggests that industry returns contain systematic components that can serve as potential common factors.

Similar to the asymptotic principal component advocated by Connor and Korajczyk (1988) to extract common factors from stock returns, we extract five industry-related common factors based on a principal component analysis.[9] The details for the construction of the industry-related factors are given in Appendix A.

This section first examines if industry-based common factors carry significant risk premiums based on Fama-MacBeth cross-sectional regression. We then examine whether it is possible to construct a better asset-pricing model with the inclusion of industry-based factors. To do so, we use the stochastic discount factor representation proposed by Cochrane (2005) along with Hansen's (1982) GMM methodology to evaluate the validity of various asset-pricing models in explaining industry returns. The famous GRS F-test is also used to examine if there are significant pricing deviations from an asset-pricing model.

4.1. Fama-MacBeth regressions

We first examine if industry-based common factors carry significant risk premiums, using Fama-MacBeth cross-sectional regressions. Following Fama and French's (1992) procedure, we estimate the "post-ranking" industry-based factor loadings for each security, then estimate the average factor risk premiums based on the Fama-MacBeth cross-sectional regressions. All securities are grouped into 192 three-way sorting portfolios on the basis of the interaction of 48 industry portfolios, 2 size groups, and 2 BM groups. In June of year t, the stocks of an industry get classified into four subportfolios based first on size and then on BM ratio. The classification relies on dependent sorting to ensure that each subportfolio has a few firms in it. However, some portfolios still have missing returns because of the smaller sample size in early years. The value-weighted monthly returns on the resulting 192 portfolios are calculated for July of year t to June of year t+1. The portfolios are rebalanced every year.

[9] Prior research suggests one to six common factors (e.g., Connor and Korajczyk, 1993; Zhou, 1999); we set the number of common factors to five (e.g., Connor and Korajczyk, 1988; Brennan, Chordia and Subrahmanyam, 1998, hereafter BCS), which we extract from the returns of the 48 industries.

To calculate the unconditional industry factor loadings for each of the 192 portfolios, we regress the portfolio returns on the five industry-based factors. Next, we assign the factor loadings to the stocks of the 192 portfolios to which they belong at the end of June of year t. To examine how these factors relate to size, BM, and momentum, we estimate the following cross-sectional regression:

$$R_{it} = \gamma_{0t} + \gamma_{At} MV_{i,t-1} + \gamma_{Bt} BM_{i,t-1} + \gamma_{Ct} P12M_{i,t-1} + \sum_{k=1}^{5} \gamma_{kt} \hat{\beta}_{ik} + \varepsilon_{it}, \tag{1}$$

where $\hat{\beta}_{ik}$ denotes stock i's factor loading on the kth industry factor loadings at time t. $MV_{i,t-1}$ and $BM_{i,t-1}$ are stock i's market capitalization and BM ratio, respectively, prior to time t-1. $P12M_{i,t-1}$ is stock i's previous twelve-month average return, calculated from time $t-14$ to time $t-2$. The one-month lag between the dependent variable R_{it} and the independent variable $P12M_{i,t-1}$ enables us to avoid potential spurious regression results caused by microstructure factors such as bid-ask bounce and non-synchronous trading.

Table 3 contains the average slopes of the monthly cross-sectional regressions and the robust t-statistics, calculated on the basis of the Newey-West (1987) adjustment for serial correlation and conditional heteroscedasticity. In addition to the full sample, we report the results for the two subperiods, with 1981 as the cut-off point again.[10]

[Insert Table 3 about here]

The premiums on the last two factors are significantly positive for both the full sample and the two subsamples (Panel A, Table 3). The average estimates of γ_4 and γ_5 are, respectively, 0.26 (t = 5.38) and 1.12 (t = 3.85), and may provide additional explanatory power beyond size, BM, and prior returns. The first factor also carries a significant factor premium, though it is significant only for the first subperiod, 1963-1981.

The coefficient of prior returns ($P12M$) is statistically insignificant when we include the five industry-based factors, which echoes the view that the momentum effect relates to

[10]The results in this section are based on unconditional tests, because we used post-ranking estimates for industry factor loadings. In unreported robustness tests, we performed the Fama-MacBeth regressions with factor loadings estimated according to the previous five-year's data. The results are qualitatively similar and available on request.

industry (Moskowitz and Grinblatt, 1999). In sharp contrast, BCS (1998) assert that the momentum effect persists even after adjusting the returns according to the Fama-French factor model or an alternative APT-based model proposed by Connor and Korajczyk (1988). Avramov and Chordia (2006) also find that the prior return effect cannot be explained by popular asset-pricing models, whether conditionally or unconditionally. Small-firm and BM premiums retain their significance regardless with the inclusion of the industry factors. The coefficient of market capitalization is -0.17 (t = -3.95), and that of BM is 0.48 (t = 8.52) for the full sample.

Panels B and C in Table 3 display the results for January and non-January months, respectively. The last two industry factors are significant in both month groupings, but the first industry factor is significant only for January during 1982-2006 and for non-January months during 1963-1981. We explore this potential seasonal component in industry returns subsequently. The small-firm effect is significant only for January months, whereas the BM effect is significant mostly for non-January months. Moreover, the coefficient of prior one-year returns ($P12M$) is significantly negative in January but significantly positive in non-January months, especially during 1963-1981.

These empirical results have two main implications. First, there exists a one-year momentum effect in non-January months, but its significance is offset by a pronounced negative momentum premium in January. This combination of effects is why we observe insignificant momentum effect over the full sample. The source of such seasonality in the momentum effect deserves further investigation. Second, some researchers argue that the small-firm effect is essentially a January effect, but our empirical findings indicate the January effect goes beyond just a January small-firm effect. The negative premium on market value in January might have been the result of previous year-end tax-loss selling, yet there is a strong return reversion in January. Thus, our empirical results reveal that industry portfolios contain additional components not fully captured by the Fama-French three factors that provide additional explanatory power for stock returns beyond size, BM, and prior returns.

4.2. *GMM tests of stochastic discount factor model*

When analyzing excess returns, it is well-known that the stochastic discount factor representation is:

$$E(R_t m_t) = 0,$$

where $R_t = (R_{1t}, ..., R_{Nt})'$ is a vector of excess returns on N assets, and m_t is the "pricing kernel." The promise of the stochastic discount factor representation, along with Hansen's GMM test, is its ability to transparently handel nonlinear or otherwise complex models (Cochrane, 2005), and has been applied extensively (e.g., Vassalou, 2003; Fletcher and Kihanda, 2005; Balvers and Huang, 2009; Aretz, Bartram, and Pope, 2010; Darrat, Li, and Parka, 2011; Chabi-Yo, 2011).

In the case of linear pricing, the pricing kernels have the following form of representation:

$$m_t = 1 - b' f_t,$$

where f_t are the models' pricing factors (see, e.g., Cochrane, 2005; Lozano and Rubio, 2011).[11] We assess the validity of an asset pricing model with Hansen's J-test, which is given as follows:

$$J \equiv T g_T(b)' S^{-1} g_T(b) \sim \chi^2(\text{number of moments - number of parameters}),$$

where T is the sample size, $g_T(b)$ stands for the vector of moment conditions evaluated at the parameter estimates, and S^{-1} is the optimal weighting matrix, estimated as the inverse of the spectral density matrix.[12]

For comparison purposes, we also examine the validity of various asset-pricing models based on the GRS test, which has the following form (Campbell, Lo and MacKinlay, 1997):

$$GRS \equiv \frac{(T - N - K)}{N} [1 + \hat{\mu}_K' \hat{\Omega}_K^{-1} \hat{\mu}_K]^{-1} \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha} \tag{2}$$

[11] While there has been a long debate concerning the relative performance of the SDF/GMM method and the traditional "Beta" method, Garrett, Hyde, and Lozano (2010) show that there appears to be a degree of trade-off between the two methods, depending on the distributional features of the factors.

[12] We are grateful to Keven Aretz for providing us his Matlab program on the estimation and test.

where $\hat{\alpha} = (\hat{\alpha}_1, ..., \hat{\alpha}_N)'$ are the OLS estimates of the intercepts, obtained by running the excess returns of the N test assets on the set of K factor portfolios. $\hat{\mu}_K$ and $\hat{\Omega}_K$ are, respectively, the maximum likelihood estimates (MLEs) of the mean and covariance matrix of the factor portfolios, and $\hat{\Sigma}$ is the MLE of the covariance matrix for the error terms. Under *iid* normality assumption and the null hypothesis of zero intercepts, the GRS statistic has a well-known central F-distribution with N and $(T - N - K)$ degrees of freedom.

4.2.1. Asset pricing models

We consider two industry-related asset-pricing models based on common factors extracted from industry returns. First, we investigate a five-factor model based solely on industry factors, which we refer to as the *industry-based five-factor model*:

$$R_{it} = \beta_{i0} + \beta_{i1}F_{1t} + \cdots + \beta_{i4}F_{4t} + \beta_{i5}F_{5t} + \varepsilon_{it}. \tag{3}$$

Second, we assess a five-factor model that augments the Fama-French three-factor model with two industry factors and takes the label *industry-augmented five-factor model*:

$$R_{it} = \alpha_i + \beta_i RMRF_t + s_i SMB_t + h_i HML_t + \beta_{i4}F_{4t} + \beta_{i5}F_{5t} + \varepsilon_{it}. \tag{4}$$

The two added industry factor portfolios are the fourth and fifth factors of the industry-based factors. We do not include the first three industry factors because the results in Table A indicate they are sufficiently explained by the Fama-French three-factor model.

For comparison purposes, we also consider three common asset-pricing models: the Sharpe-Lintner CAPM, Fama and French's (1993) three-factor model, and Carhart's (1997) momentum-augmented four-factor model. Fama and French's three-factor model is the most popular empirical multifactor model. Carhart's (1997) four-factor model augments the Fama-French three-factor model by incorporating a one-year momentum factor; empirical evidence suggests a close link between momentum and industry returns (e.g., Moskowitz and Grinblatt, 1999). The momentum factor data also come from French's Website.

For test assets, we consider three different sets of portfolios: the 25 size-BM portfolios, the 48 industry portfolios, and a larger set of portfolios combining the size-BM portfolios and industry portfolios, resulting in a total of 73 portfolios. LNS (2010) advise researchers to expand the set of test portfolios beyond size and BM portfolios because inferences based on the size-BM portfolios are misleading.[13]

4.2.2. Results

The results in Table 4 indicate that the pricing validity of the Fama-French three-factor is rejected with both tests for all sets of test assets. The CAPM is rejected with respect to the 25 size-BM portfolios and 73 size-BM-industry portfolios, but not with respect to the 48 industry portfolios. The performance of Carhart's four-factor model is slightly better than the Fama-French three model because its pricing validity holds with respect to the 25 size-BM portfolios and the 48 industry portfolios under the SDF/GMM test.

[Insert Table 4 about here]

We next examine the performance of the two industry-related models. For the 25 size-BM portfolios, the industry-augmented five-factor model is rejected for both the SDF/GMM and GRS test at the 5% significance level, but the industry-based five-factor model is not rejected by the SDF/GMM test; the J-statistic is 21.69 with a p-value of 0.36. For the 48 industry portfolios, the industry-augmented model is not rejected by the SDF/GMM test, but is rejected by the GRS test. In contrast, the industry-based model is not rejected for both tests. It seems that the industry-based five-factor model has the best pricing ability among the five models. However, when the 73 size-BM-industry portfolios serve as test assets, *all* five models are rejected with very small p-values for both the SDF/GMM and GRS tests. Thus, while the empirical results indicate that industry-related factors do have some explanatory power on stock returns, there are still significant pricing deviations uncaptured by the five asset-pricing models we consider here.

In summary, we find that the cross-sectional variation in industry returns cannot be explained fully by popular asset-pricing models. Industry-related components are not fully

[13]LNS (2010) show that the size-BM portfolios exhibit a strong factor structure because more than 90% of the time-series variation of the portfolio returns and more than 75% of the cross-sectional variation in the average return are explained by the FF model. They find that the performance of most existing models is disappointing when they implement the tests on samples expanded beyond the size-BM portfolios. They therefore suggest that researchers use larger samples, beyond the size-BM portfolios, such as expanding the size-BM portfolios with industry portfolios, to allow for more powerful inferences.

captured by existing pricing factors, perhaps because the economy is multifactorial, and industry portfolios reflect some missing factors uncaptured by common factors. Common factors extracted from industry returns entail significant factor risk premiums, which implies there may be a rational driving force of industry returns. However, attempts to construct an asset pricing model by incorporating industry-based factors cannot succeed in explaining the cross-sectional variation in average stock returns. A possibility is that industry returns reflect market mispricing, which is behavioral in nature. We explore this possibility in the next section.

5. Industries as a behavioral factor

Although industry portfolios convey additional information about stock returns, their explanatory ability appears independent of the effects of size, BM, and past returns. The failure of the risk-based explanation to account for size, BM, and momentum effects might be because firm-related anomalies are at least partly behavioral in nature (Daniel and Titman, 1997). Therefore, we also explore the role of industries from behavioral viewpoints.

We first investigate whether size, BM, and momentum premiums are stable for stocks within and across industries; some researchers argue that anomalies such as the value effect are essentially intra-industry phenomena. We also consider whether the premiums exhibit any asymmetric patterns for firms that rank above and below their industry averages on various firm characteristics. According to the prospect theory (Kahneman and Tversky, 1979), agents will be risk-taking (risk-averse) when they fall below (above) some reference point.

5.1. Within- vs. across-industry effects on size, BM, and past returns

To examine if the return regularities related to size, BM, and past returns are intra-industry phenomena, we perform a Fama-MacBeth (1973) cross-sectional regression of the following model:

$$\begin{aligned} R_i &= \gamma_0 + \gamma_1(MV_i - MV_I) + \gamma_2(MV_I - \overline{MV}) + \gamma_3(BM_i - BM_I) \\ &+ \gamma_4(BM_I - \overline{BM}) + \gamma_5(P12M_i - P12M_I) + \gamma_6(P12M_I - \overline{P12M}) + e_i, \end{aligned} \tag{5}$$

where MV_I refers to the median of market values of firms in industry I, $\overline{MV}$ is the median of MV_I across all industries, and BM_I, $\overline{BM}$, $P12M_I$ and $\overline{P12M}$ are defined accordingly.[14] To calculate an industry median, we require at least three firms in that industry. The coefficients γ_1, γ_3, and γ_5 capture the within-industry premiums, whereas the coefficients γ_2, γ_4, and γ_6 reflect across-industry premiums.

We test if the slopes of within-industry and across-industry firm characteristics are the same by examining $\gamma_1 = \gamma_2$, $\gamma_3 = \gamma_4$, and $\gamma_5 = \gamma_6$. If the within- and across-industry premiums are equal, then one may conclude that the inclusion of industry reference points is redundant. In addition to reporting the average slopes of the monthly cross-sectional regressions, we calculate the t-statistics on the basis of the Newey-West (1987) adjustment for serial correlation and heteroscedasticity.

[Insert Table 5 about here]

In Table 5, we provide the average within-industry and across-industry premiums for size, BM, and one-year past returns in various scenarios. With regard to the size premium, we note several interesting features. First, both the within- and across-industry size premiums are significantly negative; the average estimates of γ_1 and γ_2 are -0.12 (t = -2.76) and -0.20 (t = -2.42), respectively. Second, across all scenarios, the difference between γ_1 and γ_2 is insignificantly different from 0, which suggests that the size effect is not related to industry classifications. Nevertheless, the across-industry small-firm premium seems larger than the intra-industry size premium. Panels B and C of Table 5 further indicate that the small-firm premium comes from January observations, and is slightly stronger for firms across industries.

For the BM premium, Panel A suggests a generally within-industry phenomenon, because only γ_3 is highly significant. The across-industry BM premium γ_4, in contrast, is mostly insignificant with one exception: The across-industry BM effect γ_4 is -2.92 (t = -2.21) for January months during 1982-2006. Panels B and C further reveal that the BM

[14]For notational simplicity, we suppress the time index t in equations when doing so causes no confusion.

effect is significant only during 1982-2006, and it actually comes from the intra-industry BM effect in non-January months.

Recall that the results in Table 3 indicated no BM effect in January; here, the results suggest no BM effect at the individual firm level but significant negative BM effect at the industry level. That is, we find that "value industries" experience lower average returns than do "growth industries" in January. To our knowledge, the negative across-industry BM effect has not been documented previously. It is unlikely that this peculiar seasonal pattern has a risk-based explanation; instead, it probably relates to investors' year-end trading behavior, which experienced a structural change after 1981.

The last three columns of Table 5 provide the results for the one-year momentum effect. Panel A indicates that the coefficient of the across-industry past one-year return, γ_6, is significantly positive (the average estimate is 0.12, with a *t*-statistic of 2.25); therefore, the one-year momentum is essentially an across-industry effect, in line with Moskowitz and Grinblatt's (1999) finding. Panels B and C in turn show that the across-industry momentum premium is significant in non-January months. Furthermore, firms with higher past one-year returns earn lower average returns in January (both γ_5 and γ_6 are significantly negative in Panel B), which implies a return reversion in this month. The January return reversion is stronger for firms across industries after 1982 (average estimate of γ_6 is -0.71, but the estimate of γ_5 is only -0.30). Thus, the results indicate that momentum is an across-industry effect, with a reversion in January. It clearly differs from the seasonal patterns documented in prior literature.

Overall, these results show that industry classifications relate to the BM and momentum effects but not to the size effect. We find that the BM effect is basically an intra-industry phenomenon, whereas the momentum effect is an across-industry one. However, peculiar reversal patterns in January appear for both effects: a negative across-industry BM effect and a negative momentum effect. The rationales for these seasonal patterns demand further investigations.

5.2. Are premiums on size, BM, and past returns symmetric?

Fiegenbaum and Thomas (1988) and Fiegenbaum (1990) document that firms have different risk attributes when they rank above or below their industry averages on certain features. Bowman (1980) previously used accounting measures and identified a negative relationship between risk and average return; this paradoxical phenomenon is known as the "risk-return paradox" in accounting literature. Fiegenbaum and Thomas (1988) and Fiegenbaum (1990) show that the risk-return paradox reflects Kahneman and Tversky's (1979) prospect theory, which predicts that agents have different risk attitudes toward gains and losses.

Using returns on assets (ROA) as their measure for return, Fiegenbaum and Thomas (1988) and Fiegenbaum (1990) document that (1) a negative association exists between risk and return for firms whose returns are below their industry target levels, such that firms engage in risk-seeking behavior when facing loses; (2) a positive association exists for firms with returns above the industry target; and (3) the below target trade-off is generally steeper than that above the target. When the regression applies to all firms, the estimate of the slope term is dominated by the below-target firms, which have a steeper negative risk-return relationship.

Does this asymmetric risk-return relationship hold for market performance measures? Measures such as market capitalization and BM conceptually reflect investors' expectations about firms' future prospects. To the extent that markets are efficient and investors are not completely risk averse but rather have asymmetric preferences, it is possible that premiums on size, BM, and past returns also exhibit asymmetric patterns. Therefore, from an industry viewpoint, we estimate the following Fama-MacBeth (1973) cross-sectional regression:

$$\begin{aligned} R_i \;=\;& \gamma_0 + \gamma_1(MV_i - MV_I) + \gamma_2(MV_i - MV_I)I(MV_i > MV_I) + \gamma_3(BM_i - BM_I) \\ +\;& \gamma_4(BM_i - BM_I)I(BM_i > BM_I) + \gamma_5(P12M_i - P12M_I) \\ +\;& \gamma_6(P12M_i - P12M_I)I(P12M_i > P12M_I) + e_i, \end{aligned} \tag{6}$$

where $I(A)$ denotes an indicator function that takes a value of 1 when the statement A is true and 0 otherwise. The γ_2, γ_4 and γ_6 coefficients capture the asymmetric effects in size, BM, and momentum premiums, respectively. The results in Table 6 indicate three main empirical findings.

[Insert Table 6 about here]

1. Size effect. First, the small-firm effect is a below-average effect, in that the negative size-return relation exists only for firms whose market capitalization falls below their industry averages. Panel A of Table 6 reveals that γ_1 is significantly negative, whereas γ_2 is significantly positive. The average estimate is -0.26 (t = -4.76) for γ_1 and 0.25 (t = 4.25) for γ_2. This result suggests an asymmetry in size premiums for firms below and above their industry averages. Because the average size premium for firms above their industry benchmark, calculated as $\gamma_1 + \gamma_2$, is not significantly different from 0, size-return relationship does not appear to exist for above-average firms. Second, a pervasive January small-firm effect influences *all* firms. Panel B shows that the coefficients of γ_1 and $\gamma_1 + \gamma_2$ are both significantly negative in January. However, the small-firm premium is weaker for above-average firms, because the estimate of γ_2 is significantly positive. Third, the negative size-return relationship does not apply to above-average firms overall, yet Panel C implies that there is a significant positive size-return relationship for above-average firms in non-January months during 1982-2006. The estimate of $\gamma_1 + \gamma_2$ is 0.11 (t = 2.19) for non-January months during 1982-2006, which clearly is not consistent with extant risk-based explanations.
2. BM effect. The average estimate of γ_4, as reported in the fourth column of Table 6, is insignificantly different from 0 (0.07, with a t-statistic of 0.55). Therefore, there is no asymmetric pattern in the value premium overall. However, Panels B and C imply some seasonality patterns in the BM effect. In January months, below-average firms earn negative BM premiums, whereas the above-average firms earn positive BM premiums. The negative BM premium for below-average firms is stronger during 1982-2006. In addition, though the BM effect exists for all firms in non-January months, the premium is stronger for below-average firms during 1982-2006, according to the significantly negative estimate of γ_4 during this period (γ_4 = -0.34; t-statistic = -2.29).

3. Momentum effect. The estimates of γ_5 and γ_6, in the fifth and sixth columns of Table 6, are mostly insignificant, except in January. Panel B indicates that the average estimate of γ_5 is significantly negative, whereas the estimate of $\gamma_5 + \gamma_6$ is insignificant different from 0. Thus, there is no momentum effect, but there is a return reversion for firms whose returns are below the industry average in January. In other words, we find a return reversion for past losers in an industry in January, but not for past winners. The January return reversion thus is not unique to small firms but also applies to losing firms of an industry.

We thus derive two major findings. First, there is January seasonality in all effects. The January effect is not just a January small-firm effect or a small-firm January effect, and it is perhaps more complicated than prior literature has recognized. In particular, the reversal patterns differ for all three effects, which adds further complications to the nature of the asset pricing anomalies.

Second, the asymmetric pattern appears only for the small-firm effect. Conceptually, if investors are rational and market capitalization is inversely related to a firm's riskiness, the negative size-return relationship should exist for *all* firms, rather than just below-average firms. Thus risk-based theories do not provide a good explanation for our findings.

A famous explanation for the size effect, proposed by Knez and Ready (1997), suggests that a few small firms with extreme, high performances drive the effect. Knez and Ready find that the size-return relationship becomes positive if they trim the 1 percent of extreme observations from their analysis. They thus propose a "sea turtle egg" hypothesis, in that only a small proportion of eggs hatch and thrive.

We proceed a step further than Knez and Ready (1997) to determine what egg characteristics make them more likely to survive and become sea turtles. We identify firms whose market capitalization falls below their industry averages. In particular, the small-firm effect resembles the risk-return paradox in accounting literature on the following features:

1. There is a negative size-return relationship for firms whose market capitalization falls below their industry averages.

2. There is a positive size-return relationship for firms whose market capitalization is above their industry averages.
3. The negative relationship for below-average firms is stronger than the positive relationship for above-average firms.
4. Because the steeper slope of below-average firms dominates that of the above-average firms, a small-firm effect is observed when the regression analysis is applied to *all* firms.

If market capitalization is inversely related to a firm's risk propensity, these patterns contradict prospect theory. First, if the small-firm effect is driven by behaviors predicted by prospect theory, we should observe a positive (negative) size-return relationship for firms below (above) the industry averages, because smaller firms would be associated with poorer performances and higher risk, and risk-taking behavior in the face of losses would produce a negative risk-return relationship (and therefore positive size-return relationship) for underperforming firms, and vice versa. Second, we observe a steeper size-return relationship for below-average firms, which seems consistent with loss-aversion behavior in prospect theory. Again though, the sign of the size-return relationship is opposite what the theory predicts.

This discussion is based on the conventional notion that market capitalization relates negatively to risk. However, if market capitalization relates *positively* to risk, prospect theory could perfectly explain the small-firm effect. The notion that smaller firms are riskier seems an empirical observation rather than a theoretical statement. The CAPM, APT, and ICAPM do not predict any negative relationship between risk and market capitalization; the CAPM even implicitly implies that larger firms would be associated with higher systematic risks, other things being equal.[15] And yet, the CAPM seems silent about the relationship between market value and total risk. We have no clear explanation of what behavior re-

[15]In CAPM, the two-fund separation theorem suggests that every investor holds a combination of risk-free assets and a market portfolio, such that the weight of a stock in the market portfolio is proportional to its market capitalization. Because the mean-variance optimization setting also suggests that investors place greater weight on stocks with higher expected returns and smaller variances, firms with larger market capitalization should have higher expected return (and therefore higher betas), other things being equal.

ally drives the prospect-theory-like phenomenon, but the bottom line is that investors use industry averages as reference points in making decisions.[16]

The empirical results have revealed several interesting industry-related regularities in size, BM, and momentum premiums, but how is the asymmetric pattern related to the within/across pattern? In unreported tests, we examine the interaction of the two patterns by incorporating both effects into the regression model, and find that asymmetric and the within/across patterns are two independent effects, across the size, BM, and momentum categories.

6. Are industry behavioral patterns independent of the risk-based industry factors?

Even as we identify several industry-related patterns from both rational and behavioral viewpoints, we remain unsure how these patterns interrelate. To examine their interactions, we adopt a methodology proposed by BCS (1998) and jointly consider all effects together.

The BCS approach has several advantages over the classical Fama-MacBeth (1973) cross-sectional regression. First, it does not suffer from the errors-in-variables (EIV) problem, because the *estimated* factor loadings or market betas are not the independent variables. The EIV problem, as widely documented in asset pricing literature, states that the time-series estimation of the market beta or factor loading in the two-stage method may suffer from the measurement errors. The coefficients estimated in the second stage cross-sectional regression thus are biased toward 0. Following Fama and MacBeth (1973), most studies group firms into portfolios to minimize the EIV problem, reasoning that if the errors in the βs for different firms are not perfectly correlated, the βs of the portfolios could provide more precise estimates of true β than those for individual securities due to noise cancelation.

Second, the implementation of the BCS methodology uses single securities and thus avoids the data-snooping biases inherent to portfolio-based approaches (Lo and MacKinlay,

[16]We also explore if the negative relation is related to Fiegenbaum’s (1990) findings by incorporating ROA into the regression models. The results indicate that including ROA does not explain any of the results we find in this article. These results are available on request.

1990). Without relying on the portfolio grouping procedure, the BCS approach retains information from individual securities. As a result, the BCS approach is theoretically more powerful statistically, because a larger sample appears in the second-pass cross-sectional regression. Third, the BCS methodology can test a wide array of asset pricing models, both conditionally and unconditionally (see Avramov and Chordia, 2006).

Following BCS (1998), we examine whether the behavioral patterns we have identified can be explained by industry-related factor asset pricing models. Suppose that returns are generated by an L-factor version of the APT:

$$\tilde{R}_{it} = E\left(\tilde{R}_{it}\right) + \sum_{k=1}^{L} \beta_{ik} \tilde{f}_{kt} + \tilde{e}_{it}, \tag{7}$$

where the expected return is:

$$E\left(\tilde{R}_{it}\right) = R_{Ft} + \sum_{k=1}^{L} \lambda_{kt} \beta_{ik}. \tag{8}$$

In this setting, $\tilde{R}_{it}$ is the return on security i, R_{Ft} is the risk-free interest rate, β_{ik} is the unconditional loading of security i on factor k, and λ_{kt} is the risk premium associated with factor k. A standard application of the Fama-MacBeth (1973) procedure would involve estimating of the following cross-sectional regression:

$$\tilde{R}_{it} - R_{Ft} = \gamma_{0t} + \sum_{k=1}^{L} \lambda_{kt} \beta_{ik} + \sum_{m=1}^{M} \gamma_{mt} Z_{mit} + \tilde{e}_{it}, \tag{9}$$

where Z_{mit} $(m = 1, 2, \cdots, M)$ is the value of the (non-risk) characteristic m for security i in month t, and γ_{mt} is the premium per unit of characteristic m in month t. For this study, Z_{mit} are the industry-related firm characteristics used in the previous section. According to the null hypothesis that expected returns depend only on the factor loadings β_{ik}, the coefficients γ_{mt} $(m = 0, 1, \cdots, M)$ will be equal to 0.

As we noted previously, the standard Fama-MacBeth approach suffers from an EIV problem when testing Equation (9), because the estimations of factor loadings are subject to estimation errors. As a result, the ordinary least squares (OLS) estimates of the risk premiums associated with the factors (λ_{kt}s) and the coefficients associated with the characteristics (γ_{mt}s) are biased and inconsistent. In a simple approach to correct the bias, BCS

(1998) suggest estimating for each year the factor loadings β_{ik} for all securities that have at least 24 return observations over the previous 60 months. The estimation of the factor loadings relies on a procedure proposed by Dimson (1979), which includes one-period lag factors in the time-series regression to reduce the estimation problem due to nonsynchronous trading. The estimated risk-adjusted return on each of the securities, $\tilde{R}^*_{it}$, for each month of the following year is then:

$$\tilde{R}^*_{it} \equiv \tilde{R}_{it} - R_{Ft} - \sum_{k=1}^{L} \hat{\beta}_{ik} \tilde{F}_{kt}, \tag{10}$$

for all i, where $\tilde{F}_{kt} \equiv \lambda_{kt} + \tilde{f}_{kt}$ is the sum of the factor realization and its associated risk premium at month t. The risk-adjusted returns from Equation (10) serve to test whether non-risk firm characteristics can describe the cross-section of expected returns. The estimation equation becomes:

$$\tilde{R}^*_{it} = \gamma_{0t} + \sum_{m=1}^{M} \gamma_{mt} Z_{mit} + \tilde{e}'_{it}, \tag{11}$$

for all t. Because the measurement errors in the estimates of factor loadings transfer to the dependent variable, the OLS estimates of $\hat{\gamma}_{mt}$ $(m = 0, 1, \cdots, M)$ from Equation (10) are unbiased. For our purposes, we estimate the following cross-sectional regression:

$$\begin{aligned}
\tilde{R}^*_i \quad = \quad & \gamma_0 + \gamma_1(MV_i - MV_I) + \gamma_2(MV_I - \overline{MV}) + \gamma_3(MV_i - MV_I)I(MV_i > MV_I) \\
+ \quad & \gamma_4(BM_i - BM_I) + \gamma_5(BM_I - \overline{BM}) + \gamma_6(BM_i - BM_I)I(BM_i > BM_I) \\
+ \quad & \gamma_7(P12M_i - P12M_I) + \gamma_8(P12M_I - \overline{P12M}) \\
+ \quad & \gamma_9(P12M_i - P12M_I)I(P12M_i > P12M_I) + e_i,
\end{aligned} \tag{12}$$

where $\tilde{R}^*_i$ is the risk-adjusted return. We use the two industry-adjusted five-factor models from Section 4 as benchmark asset-pricing models.

For each month, we regress the risk-adjusted returns on the nine characteristics. Next, we compute the time-series averages of the coefficients associated with the characteristics, $\hat{\gamma}_j$ for $j = 0, 1, \cdots, 9$, and test them with the Newey-West adjustment. If industry-related factor models can explain expected stock returns, the coefficients $\hat{\gamma}_j$ should all be statistically indistinguishable from 0.

[Insert Table 7 about here]

In Table 7, the dependent variable is the returns on individual stocks adjusted according to the industry-based five-factor model (Panel A) and the industry-augmented five-factor model (Panel B).

The below-industry small-firm effect is significant regardless of the asset pricing model used to adjust for returns. For example, Panel A indicates that the average estimates of γ_1 and γ_2 are -0.48 (t = -6.99) and -0.26 (t = -4.10), respectively, which means that the small-firm effect is evident both within and across industries, though the former is stronger. The coefficient of the above-industry incremental effect, γ_3, is 0.45 (t = 6.31), which is close to the estimate of γ_1 in absolute value and thus implies there is no size effect for above-industry firms. The below-industry and across-industry size premiums are stronger in January than in non-January months. The results for the industry-augmented five-factor model, in Panel B, are close to those in Panel A.

The BM effect remains mostly a within-industry phenomenon, regardless of the asset pricing model. As in Table 5, the within-industry BM effect comes from non-January months. The above-industry BM effect in January disappears though, replaced by a negative across-industry BM premium when the returns are adjusted according to the industry-augmented five-factor model. In Panel B, this effect appears to be caused by the strong negative BM premium in January. Thus, the industry-based five-factor model seems to perform better in terms of explaining the BM anomaly. However, both models fail to account for the within-industry BM effect observed in non-January months.

Finally, the results in Table 7 indicate that the momentum effect completely disappears when we consider industry-related risk factors. However, by splitting the observations into January and non-January observations, we still identify a below-industry momentum effect in non-January months, and the effect is reversed in January. Above-industry firms do not experience significant return reversals in January. Thus, the results are the same as those reported in the previous section.

Overall, the behavioral patterns in size, BM, and momentum effects identified in the previous section thus remain mostly intact, regardless of the models used to calculate the risk-adjusted returns.

7. Some robustness checks

7.1. Industry concentration premium

Hou and Robinson (2006) find that firms in concentrated industries earn lower returns and attribute the "concentration" premium to rationality, because firms in highly concentrated industries face higher barriers to entry and therefore engage in less innovation. Such firms also are less risky and command lower expected returns. However, because their analysis relies on the Herfindahl index, calculated at the industry level, the concentration premium is an across-industry premium and probably is not a source of intra-industry and asymmetric patterns. It is nevertheless of interest to examine how the concentration premium may interact with industry-related patterns.

Therefore, we examine the robustness of the patterns we identify, as well as how they interact with the explanatory ability of industry concentration. Hou and Robinson's (2006) industry classification uses three-digit SIC codes and excludes regulated industries. Following their research design, we identify an average of 272.2 industries. In total, there are 401 industries in the full sample.

[Insert Table 8 about here]

As we show in Table 8, the patterns we have identified are robust to different industry classifications and to the inclusion of industry concentration. We also find that the industry concentration premium is sensitive to the classification scheme. When firms are classified into 48 industries, the industry concentration premium becomes insignificant.

7.2. Are these patterns rational?

If the patterns identified are rationally driven, the premiums should be related to business cycles. We collect the data on economic growth, based on the logarithmic difference of the GDP, to define the states of the economy. Following Liew and Vassalou (2000), we use quarterly observations and associate the next year's growth in GDP with the past year's annual return. We then sort by growth in GDP every quarter. "Good states" of the economy are those with the highest 25% of future GDP growth, "bad states" are those with the lowest 25% of future GDP growth, and "neutral states" comprise the remaining 50% of observations.

Liew and Vassalou (2000) emphasize that if high returns in HML and SMB are associated with future good states of the economy, high BM and small capitalization stocks are better able to prosper than are low BM and big capitalization stocks in periods of high economic growth expectations. Using 10 international markets, Liew and Vassalou (2000, table 4) find that for most countries, the premiums on SMB and HML relate positively to future economic growth. But they also find that in the U.S. market, only SMB has significantly different premiums across good and bad states among the three well-known factors (i.e., SMB, HML, and MOM).

Using the BCS regression, we examine if our identified patterns relate to the state of the economy, whether the returns are unadjusted or adjusted with the industry-related factors.

[Insert Table 9 about here]

We begin with the small-firm effect. Table 9 indicates that the below-industry size premium is significant only in good states, with or without risk adjustment. Panel A also indicates that the BM effect is still an intra-industry one, but it is slightly larger in bad states than in good states. When returns are adjusted by industry-related factors, the results are less consistent though, because the intra-industry premium (i.e., γ_4) becomes larger in good states (see Panels B and C). For the one-year momentum effect, Table 9 reveals an intra-industry momentum effect, but only in bad states.

Overall, the results indicate that the patterns identified in this article do not appear to have consistent cyclical patterns. That is, the patterns are not completely driven by rational behavior.

8. Conclusion

Although prior literature has identified several industry-related return patterns, it remains unclear how and why industries affect asset prices. In particular, we do not know how industries interact with firm characteristics such as size, BM, and past returns to explain the cross-section of stock returns.

We have explored this issue from both rational and behavioral perspectives, focusing especially on the role of industries. With our extensive CRSP/COMPUSTAT merged sample, we have uncovered several unique industry-related regularities.

First, we identified two common factors, extracted from returns on industry portfolios, that bear significant factor risk premiums. The results are consistent with literature that states that information about industries helps predict stock markets and economic fundamentals. Second, the small-firm effect is essentially a below-industry phenomenon, in that a negative size-return association exists only for firms whose market capitalization is below their industry median. Third, the value effect is an intra-industry phenomenon that is especially strong in non-January months. Fourth, the one-year momentum effect is an inter-industry phenomenon, but its significance disappears when returns are risk adjusted with industry-related factor models. A closer inspection also reveals that after risk adjustment, there is a significant below-industry momentum in non-January months and a strong return reversal for below-industry firms in January.

The empirical results further reveal that the asset pricing anomalies, including the small-firm effect, the BM effect, and the momentum effect, relate to industry classifications. Research traditionally has treated these anomalies as if they share the same nature (or driving force). Risk-based advocates view them as variables reflecting sensitivities to common factors (e.g., Fama and French, 1993, 1995; Carhart, 1997); behavioral supporters view such price-related regularities as the results of investor under- or over-reactions (e.g., Lakonishok, Shleifer, and Vishny, 1994; Daniel, Hirshleifer and Subrahmanyam, 1998). Our results indicate that size, BM, and momentum anomalies instead may have different roots.

Our empirical findings reveal that industries play a dual role with both rational and behavioral components. This finding is consistent with some views that suggest stock returns reflect both rational (i.e., covariance risk) and behavioral (i.e., mispricing) components (e.g., Daniel, Hirshleifer, and Subrahmanyam, 2002). A good asset pricing model, if it exists, must consider not only rational and mispricing components, but also their interactions with industry classifications.

Although we have identified several interesting industry-related regularities, the findings seem to raise even more questions. Why do stocks with above or below industry-average firm characteristics experience different firm characteristic-return relations? How are industry-related patterns related to the statistical cross-moments (e.g., the lead-lag relationship), and how and why are they related to pricing? If the asymmetric industry-related patterns are behavioral in essence, what are the behavioral foundations behind these relations? These topics seem worthy of further research, because a better understanding of such important phenomena may help us approach the ultimate goal of developing a unified asset pricing theory that mirrors the real world.

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Appendix A: Extracting common factors from industry returns

Recall that in Ross's (1976) APT, asset returns are assumed to be generated or "governed" by a linear factor model of the form:

$$r_i = \mu_i + \sum_{j=1}^{k} \beta_{ij} f_j + \varepsilon_i, \tag{A.1}$$

where f_j are common factors, and the idiosyncratic components ε_i are assumed to be uncorrelated across securities. Determining the number of common factors is an important empirical issue (e.g., Connor and Korajczyk, 1993; Zhou, 1999). Conceptually, a certain number of common factors is chosen to ensure that the idiosyncratic components are approximately uncorrelated across securities.

We use principal components analysis to extract common factors from the returns on industry portfolios. Specifically, let $R_I = (R_1, R_2, \cdots, R_M)$ denote a $(T \times M)$ matrix of returns of the industry portfolios, where T refers to the number of time-series observations and M is to the number of industry portfolios; R_i is a column vector of asset returns of industry i in excess of the risk-free rate. Let P denote the orthogonal $(M \times M)$ matrix whose columns p_i are characteristic vectors of $C \equiv R_I' R_I$. Then the k (here $k = 5$) factors can be constructed as

$$f_j = R_I\, p_j, \quad j = 1, \cdots, k, \tag{A.2}$$

where f_j is a $(T \times 1)$ vector that refers to the jth factor extracted from the industry portfolios.

The principal component f_j is not yet a portfolio, because the "weights" p_j on the industry portfolios do not sum to 1. With a minor adjustment in weights, the excess factor return on the jth industry factor F_j is as follows:

$$F_j = R_I\, \hat{p}_j, \quad j = 1, \cdots, k, \tag{A.3}$$

where $\hat{p}_j = \frac{p_j}{\sum_{l=1}^{M} p_{jl}}$ is normalized portfolio weights.

The second and third columns of Table A report the marginal and cumulative proportions of variations in industry returns explained by the five industry factors. The first factor alone captures 58% of the variation in industry returns, and in total, the five factors explain 73% of the variation in industry returns.

[Insert Table A about here]

We also regress the returns on each of the five factors on the Fama-French three-factor model and Carhart's four-factor model to see how they interact and uncover some interesting findings. First, the first extracted factor has a high adjusted R^2 of 0.93 and loads heavily on all four factors, including the market portfolio, SMB, HML, and momentum (MOM) factors. Second, though the five industry-based factors relate significantly to all four factors, only the last two industry-based factors have significant intercepts. The t-values of the intercept of the two last factors are 5.18 and 2.24 under the Fama-French three-factor model, which suggests that the two industry-based factors contain significant components uncaptured by the Fama-French factors. The intercept of the fourth factor F_4 remains significantly different from zero under the Carhart four-factor model (t = 5.10). Overall, these results are consistent with Fama and French's (1997) finding that the three-factor model fails to provide good estimates of the industry cost of equity.

Table 1: Descriptive and Summary Statistics of 48 Industry Portfolios

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. The calculation is based on the maximum available time-series length in each industry. The average raw return is the time-series average of value-weighted returns in each industry. The standard deviation of raw returns is the time-series standard deviation of raw returns in each industry. The abnormal return is the adjusted return using the three-factor model. The average ln(ME) is the time-series average of total market capitalization in each industry. The average BM is the time-series average of book-to-market equity in each industry. The average number of firms is the time-series average of firms in each industry (numbers in parentheses are standard deviations). In each June of year *t*, BM was calculated from the book value of fiscal year *t*-1 to the market equity at the end of year *t*-1. The book value is stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. We include firms have been on COMPUSTAT for at least two years and exclude firms with negative book values. The F-statistic corresponds to the GRS test for the joint hypothesis that all abnormal returns are zero. As in Fama and French (1992), the smallest and largest 0.5% of the observations on BM are set to the next smallest and largest value of the variable.

	Industry	Avg. Raw Return (%)	S.d. of Raw Return (%)	Abnormal Return (%)	Avg. ln(ME)	Avg. BM	Avg. No. of Firms
1	Agriculture	1.23	6.11	0.10	8.41	0.44	11.6 (5.0)
2	Food Products	1.12	4.44	0.17	11.23	0.42	81.5 (16.6)
3	Candy and Soda	1.38	7.44	0.06	8.68	0.37	7.5 (3.2)
4	Beer and Liquor	1.24	5.17	0.40	10.63	0.38	14.6 (3.8)
5	Tobacco Products	1.62	6.42	0.72	9.94	0.29	5.9 (2.1)
6	Recreation	0.87	7.72	-0.53	9.06	0.43	37.4 (9.3)
7	Entertainment	1.28	6.96	-0.02	9.98	0.37	51.3 (26.3)
8	Printing and Publishing	1.11	5.68	-0.01	10.46	0.31	40.6 (13.9)
9	Consumer Goods	0.97	4.81	0.15	11.18	0.36	86.7 (24.4)
10	Apparel	1.14	6.53	-0.21	9.79	0.51	68.6 (16.8)
11	Healthcare	1.24	8.83	-0.15	10.16	0.32	50.5 (39.4)
12	Medical Equipment	1.18	5.75	0.32	10.75	0.28	93.7 (64.2)
13	Pharmaceutical Products	1.24	5.18	0.64	12.10	0.21	124.8 (105.0)
14	Chemicals	0.99	5.35	-0.17	11.32	0.37	73.9 (12.2)
15	Rubber and Plastic Products	1.05	5.96	-0.19	9.16	0.45	49.0 (15.2)
16	Textiles	1.01	6.22	-0.40	8.72	0.62	41.3 (16.1)
17	Construction Materials	1.05	5.53	-0.24	10.55	0.44	112.4 (33.6)
18	Construction	0.98	7.32	-0.50	9.45	0.49	47.8 (21.6)
19	Steel Works Etc	0.94	6.70	-0.47	10.39	0.48	73.7 (12.4)
20	Fabricated Products	0.73	7.24	-0.83	7.68	0.59	21.9 (7.5)

Table 1 Continued

	Industry	Avg. Raw Return (%)	S.d. of Raw Return (%)	Abnormal Return (%)	Avg. ln(ME)	Avg. BM	Avg. No. of Firms
21	Machinery	1.05	6.10	-0.10	11.10	0.40	149.2 (34.2)
22	Electrical Equipment	1.06	5.72	-0.09	10.16	0.38	64.6 (18.5)
23	Automobiles and Trucks	0.86	6.09	-0.47	11.14	0.45	65.8 (10.4)
24	Aircraft	1.20	6.52	-0.09	10.63	0.51	27.0 (5.3)
25	Shipbuilding, Railroad Equip.	1.17	7.41	-0.25	9.03	0.49	11.9 (2.3)
26	Defense	1.22	7.08	-0.08	9.06	0.47	6.3 (1.8)
27	Precious Metals	1.11	9.79	0.01	8.83	0.23	14.6 (8.2)
28	Non-Metallic Mining.	1.34	6.41	0.05	8.88	0.37	19.4 (5.8)
29	Coal	1.33	8.70	0.01	8.16	0.40	6.1 (1.6)
30	Petroleum and Natural Gas	1.20	5.58	0.16	12.16	0.36	157.0 (60.7)
31	Utilities	0.89	4.11	-0.15	12.11	0.54	158.5 (5.5)
32	Communication	0.88	4.88	0.03	12.21	0.29	80.2 (21.9)
33	Personal Services	1.09	6.62	-0.16	9.68	0.38	37.2 (18.2)
34	Business Services	0.99	6.16	0.28	12.31	0.32	289.0 (212.4)
35	Computers	1.11	8.00	0.31	11.49	0.29	132.0 (85.9)
36	Electronic Equipment	1.18	7.96	0.23	11.81	0.37	192.0 (89.3)
37	Measuring and Control Equip.	1.11	7.55	-0.04	10.18	0.41	81.2 (39.3)
38	Business Supplies	0.97	5.26	-0.12	11.10	0.60	61.8 (18.2)
39	Shipping Containers	1.13	6.05	-0.05	8.90	0.44	15.8 (4.9)
40	Transportation	1.02	5.95	-0.22	11.09	0.41	91.7 (9.4)
41	Wholesale	0.92	5.58	-0.27	10.78	0.48	147.5 (56.1)
42	Retail	1.10	5.63	0.02	12.11	0.37	217.0 (59.8)
43	Restaraunts, Hotels, Motels	1.31	6.75	0.14	10.32	0.35	65.9 (34.8)
44	Banking	1.08	5.59	-0.16	12.59	0.31	388.7 (277.3)
45	Insurance	1.17	5.73	0.00	11.97	0.27	129.6 (73.3)
46	Real Estate	0.79	6.90	-0.79	8.17	0.39	40.3 (14.4)
47	Trading	1.46	6.84	0.18	11.16	0.16	107.8 (51.1)
48	Miscellaneous	1.23	8.97	-0.24	9.78	0.49	22.0 (19.1)
	Average	1.11	6.39	-0.06	10.36	0.40	82.0
	F-statistic (all = 0)			2.29			
	(p-value)			(< 0.01)			

Table 2: Within- and Across-Industry Fama-French Risk-Adjusted Return Correlations

For the full sample period, July 1963 to December 2006, we classify each individual firm into a specific industry according to SIC codes from COMPUSTAT. Industry classifications are based on 5 to 48 industries, defined by French, and two- to four-digit SIC codes. For each stock, we first regress their returns on the Fama-French three-factor model, and use the residuals to estimate the full-period correlations for each pair of individual firms over various intervals k. This table reports within-industry correlations for firms that belong to the same industry, and across-industry correlations for firms that belong to different industries.

Classification	Number of industries	k = 1	2	3	4	5	6	7	8	9	10	11	12
Panel A: Within-Industry Correlations													
5 industries	5	0.0165	0.0178	0.0224	0.0183	0.0267	0.0242	0.0351	0.0350	0.0456	0.0528	0.0180	0.0194
10 industries	10	0.0252	0.0195	0.0318	0.0285	0.0432	0.0425	0.0554	0.0676	0.0196	0.0210	0.0279	0.0215
12 industries	12	0.0355	0.0320	0.0493	0.0483	0.0629	0.0786	0.0203	0.0219	0.0294	0.0223	0.0379	0.0341
17 industries	17	0.0531	0.0520	0.0674	0.0857	0.0228	0.0245	0.0326	0.0251	0.0417	0.0378	0.0585	0.0573
30 industries	30	0.0736	0.0931	0.0231	0.0249	0.0330	0.0253	0.0429	0.0389	0.0610	0.0592	0.0762	0.0990
38 industries	38	0.0245	0.0263	0.0345	0.0266	0.0446	0.0405	0.0637	0.0616	0.0792	0.1025	0.0251	0.0271
48 industries	48	0.0353	0.0272	0.0462	0.0416	0.0654	0.0635	0.0814	0.1062	0.0265	0.0286	0.0370	0.0292
2-digit	67	0.0485	0.0439	0.0688	0.0664	0.0854	0.1112	0.0271	0.0292	0.0374	0.0292	0.0491	0.0443
3-digit	261	0.0695	0.0672	0.0861	0.1138	0.0287	0.0310	0.0391	0.0306	0.0510	0.0463	0.0719	0.0696
4-digit	410	0.0880	0.1157	0.0293	0.0316	0.0401	0.0311	0.0523	0.0474	0.0737	0.0706	0.0902	0.1201
Panel B: Across-Industry Correlations													
5 industries	5	0.0055	0.0058	0.0058	0.0056	0.0060	0.0062	0.0064	0.0066	0.0071	0.0074	0.0031	0.0035
10 industries	10	0.0036	0.0035	0.0038	0.0041	0.0044	0.0045	0.0053	0.0057	0.0026	0.0031	0.0031	0.0031
12 industries	12	0.0034	0.0037	0.0040	0.0042	0.0051	0.0055	0.0017	0.0022	0.0022	0.0022	0.0025	0.0028
17 industries	17	0.0032	0.0034	0.0044	0.0048	0.0026	0.0032	0.0032	0.0031	0.0034	0.0038	0.0042	0.0044
30 industries	30	0.0055	0.0060	0.0018	0.0024	0.0025	0.0024	0.0027	0.0031	0.0035	0.0037	0.0049	0.0054
38 industries	38	0.0025	0.0031	0.0033	0.0032	0.0035	0.0038	0.0042	0.0045	0.0057	0.0062	0.0022	0.0028
48 industries	48	0.0030	0.0029	0.0032	0.0036	0.0040	0.0043	0.0056	0.0061	0.0029	0.0035	0.0038	0.0036
2-digit	67	0.0039	0.0043	0.0048	0.0051	0.0064	0.0069	0.0029	0.0035	0.0038	0.0037	0.0040	0.0044
3-digit	261	0.0049	0.0051	0.0065	0.0069	0.0035	0.0042	0.0046	0.0045	0.0048	0.0051	0.0056	0.0059
4-digit	410	0.0073	0.0078	0.0031	0.0039	0.0042	0.0041	0.0044	0.0048	0.0053	0.0057	0.0071	0.0076
Panel C: Difference between Within- and Across-Industry Correlations													
5 industries	5	0.0110	0.0120	0.0166	0.0127	0.0207	0.0180	0.0287	0.0285	0.0385	0.0454	0.0149	0.0159
10 industries	10	0.0216	0.0160	0.0280	0.0244	0.0388	0.0379	0.0501	0.0620	0.0170	0.0178	0.0248	0.0184
12 industries	12	0.0321	0.0283	0.0453	0.0441	0.0578	0.0731	0.0186	0.0197	0.0272	0.0201	0.0354	0.0313
17 industries	17	0.0499	0.0486	0.0630	0.0808	0.0202	0.0213	0.0294	0.0220	0.0382	0.0341	0.0543	0.0529
30 industries	30	0.0680	0.0871	0.0214	0.0225	0.0305	0.0229	0.0402	0.0358	0.0575	0.0555	0.0713	0.0936
38 industries	38	0.0221	0.0232	0.0313	0.0234	0.0411	0.0366	0.0595	0.0570	0.0734	0.0963	0.0229	0.0242
48 industries	48	0.0323	0.0243	0.0429	0.0380	0.0614	0.0591	0.0759	0.1002	0.0237	0.0251	0.0332	0.0256
2-digit	67	0.0446	0.0396	0.0641	0.0614	0.0790	0.1043	0.0242	0.0257	0.0336	0.0254	0.0451	0.0399
3-digit	261	0.0647	0.0620	0.0797	0.1069	0.0252	0.0268	0.0345	0.0261	0.0462	0.0411	0.0663	0.0637
4-digit	410	0.0807	0.1080	0.0262	0.0277	0.0359	0.0270	0.0479	0.0426	0.0684	0.0649	0.0831	0.1125

Table 3: Average Slopes of Size, BM, Past Returns, and Industry Factors

We estimate the factor loadings of stocks using the full period sample in 192 portfolios. In June of year t, the NYSE, AMEX, and NASDAQ stocks that meet the CRSP-COMPUSTAT data requirements are allocated to two size portfolios in each industry using the NYSE size breakpoints. The NYSE, AMEX, and NASDAQ stocks in each size portfolio of each industry are then sorted into two BM portfolios using the BM for year t-1. The value-weighted monthly returns on the resulting 192 portfolios are then calculated for July of year t to June of year t+1. The industry factor loadings β_{fp} calculated from using the full period sample of post-ranking returns for each portfolio to regress on the five common factors. Stocks are assigned the industry factor loadings of one of the 192 portfolios at the end of June of year t. We estimate the following cross-sectional regressions:

$$R_{it} = \gamma_{0t} + \gamma_{At}MV_{i,t-1} + \gamma_{Bt}BM_{i,t-1} + \gamma_{Ct}P12M_{i,t-1} + \sum_{k=1}^{5}\gamma_{ft}\beta_{ik} + \varepsilon_{it},$$

where β_{ik} denotes the industry factor loadings of factor f of 192 industry portfolios to which stock i belongs at time t. The average slopes of the monthly cross-sectional regressions are reported; in parenthesis are the Newey-West (1987) t-statistics, adjusted for serial correlation and heteroscedasticity. Panels A, B, and C report the results of various subperiods for all months, January, and non-January, respectively.

	γ_A	γ_B	γ_C	γ_1	γ_2	γ_3	γ_4	γ_5
Panel A: All Months								
1963/07 – 2006/12	-0.17	0.48	0.00	0.02	-0.01	-0.20	0.26	1.12
	(-3.95)	(8.52)	(0.21)	(2.91)	(-0.03)	(-1.45)	(5.38)	(3.85)
1963/07 – 1981/12	-0.18	0.37	0.02	0.03	0.56	-0.18	0.24	0.67
	(-2.75)	(3.77)	(0.99)	(2.58)	(1.51)	(-1.56)	(3.93)	(2.34)
1982/01 – 2006/12	-0.16	0.56	-0.01	0.01	-0.43	-0.22	0.28	1.45
	(-2.79)	(8.83)	(-0.95)	(1.58)	(-1.16)	(-0.99)	(3.99)	(3.21)
Panel B: January								
1963/07 – 2006/12	-1.57	0.57	-0.34	0.08	0.58	-1.40	0.47	2.35
	(-7.50)	(1.77)	(-4.76)	(2.75)	(0.56)	(-2.77)	(2.27)	(2.56)
1963/07 – 1981/12	-1.59	1.10	-0.40	0.07	2.47	-0.65	0.03	1.33
	(-6.42)	(2.58)	(-2.86)	(1.43)	(1.68)	(-1.51)	(0.12)	(1.14)
1982/01 – 2006/12	-1.55	0.18	-0.29	0.08	-0.78	-1.93	0.78	3.08
	(-5.04)	(0.43)	(-4.69)	(2.72)	(-0.58)	(-2.50)	(2.81)	(2.29)
Panel C: Non−January								
1963/07 – 2006/12	-0.04	0.47	0.03	0.02	-0.06	-0.09	0.24	1.01
	(-1.05)	(8.28)	(2.63)	(2.23)	(-0.22)	(-0.63)	(4.70)	(3.47)
1963/07 – 1981/12	-0.05	0.30	0.06	0.03	0.39	-0.13	0.26	0.62
	(-0.84)	(3.28)	(2.69)	(2.33)	(1.12)	(-1.18)	(3.89)	(2.00)
1982/01 – 2006/12	-0.04	0.59	0.01	0.01	-0.40	-0.06	0.23	1.30
	(-0.67)	(9.00)	(0.94)	(0.88)	(-1.00)	(-0.26)	(3.21)	(2.91)

Table 4: SDF/GMM and GRS Tests on Asset Pricing Models

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. We downloaded the value-weighted returns for the 25 size/BM portfolios from Kenneth French's website. We report the J-statistic of the SDF/GMM test, and the F-statistic of GRS (1989) test. Numbers in parentheses are p-values.

	25 size-BM		48 ind.		25 size-BM+48 ind.	
Model	SDF/GMM	GRS	SDF/GMM	GRS	SDF/GMM	GRS
CAPM	93.13	4.47	52.34	1.27	162.02	2.95
	(0.00)	(0.00)	(0.27)	(0.11)	(0.00)	(0.00)
Fama-French	67.05	3.16	46.72	2.30	144.87	2.45
three-factor model	(0.00)	(0.00)	(0.40)	(0.00)	(0.00)	(0.00)
Carhart four-factor	27.43	2.54	43.44	2.36	131.31	2.19
model	(0.16)	(0.00)	(0.50)	(0.00)	(0.00)	(0.00)
Industry-based	21.69	4.94	40.55	1.19	152.41	2.74
five-factor model	(0.36)	(0.00)	(0.58)	(0.18)	(0.00)	(0.00)
Industry-augmented	34.09	2.51	41.67	1.46	119.16	1.85
five-factor model	(0.03)	(0.00)	(0.53)	(0.03)	(0.00)	(0.00)

Table 5: Average Slopes of Within- and Across-Industry Components of Size, BM, and Past Returns

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. In each June of year t, we formed BM for July of year t to June of year t+1 as the book value of the prior fiscal year divided by the market equity of the prior calendar yearend. The book value is defined as stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. As in Fama and French (1992), the smallest and largest 0.5% of the BM observations are set to the next smallest or largest value of the variable. We use a firm's market equity in June of year t to measure its size for July of year t to June of year t+1. We exclude firms with negative book value and those that have not appeared in on COMPUSTAT for at least two years. We employ the Fama-MacBeth (1973) procedure to estimate the following equation:

$$\begin{aligned} R_{it} &= \gamma_0 + \gamma_1(MV_{it} - MV_{It}) + \gamma_2(MV_{It} - \overline{MV_t}) + \gamma_3(BM_{it} - BM_{It}) \\ &+ \gamma_4(BM_{It} - \overline{BM_t}) + \gamma_5(P12M_{it} - P12M_{It}) + \gamma_6(P12M_{It} - \overline{P12M_t}) + e_{it}, \end{aligned}$$

where MV_{It} refers to the median of the market values of firms in industry I in month t; $\overline{MV_t}$ refers to the median of MV_{It} in month t; and BM_{It}, $\overline{BM_t}$, $P12M_{It}$, and $\overline{P12M_t}$ are defined accordingly. We require the industry median to be calculated with a minimum of three firms. The average slopes of the monthly cross-sectional regressions are reported; in the parenthesis, we provide the Newey-West (1987) t-statistics adjusted for serial correlation and heteroscedasticity. Panels A, B, and C report the results of various subperiods for all months, January, and non-January, respectively.

	γ_1	γ_2	$\gamma_1 - \gamma_2$	γ_3	γ_4	$\gamma_3 - \gamma_4$	γ_5	γ_6	$\gamma_5 - \gamma_6$
Panel A: All months									
1963/07 – 2006/12	-0.12	-0.20	0.08	0.33	-0.06	0.39	-0.01	0.12	-0.13
	(-2.76)	(-2.42)	(1.45)	(5.21)	(-0.30)	(2.29)	(-1.00)	(2.25)	(-2.77)
1963/07 – 1981/12	-0.14	-0.22	0.09	0.23	-0.10	0.33	0.00	0.11	-0.11
	(-1.90)	(-1.62)	(0.98)	(2.12)	(-0.43)	(1.74)	(-0.02)	(1.57)	(-1.89)
1982/01 – 2006/12	-0.12	-0.19	0.07	0.40	-0.03	0.43	-0.02	0.12	-0.14
	(-1.96)	(-1.88)	(1.11)	(5.53)	(-0.11)	(1.69)	(-1.57)	(1.66)	(-2.13)
Panel B: January									
1963/07 – 2006/12	-1.56	-1.93	0.36	0.14	-1.47	1.61	-0.32	-0.49	0.17
	(-6.87)	(-5.56)	(1.68)	(0.41)	(-1.64)	(2.40)	(-4.34)	(-2.45)	(1.03)
1963/07 – 1981/12	-1.68	-2.20	0.52	0.82	0.55	0.27	-0.36	-0.19	-0.17
	(-6.14)	(-4.62)	(1.29)	(1.71)	(0.77)	(0.41)	(-2.38)	(-0.52)	(-0.61)
1982/01 – 2006/12	-1.48	-1.73	0.25	-0.34	-2.92	2.58	-0.30	-0.71	0.41
	(-4.48)	(-3.73)	(1.09)	(-0.79)	(-2.21)	(2.63)	(-4.75)	(-3.42)	(2.40)
Panel C: Non-January									
1963/07 – 2006/12	0.00	-0.05	0.05	0.34	0.06	0.28	0.02	0.17	-0.16
	(0.11)	(-0.61)	(0.99)	(5.41)	(0.29)	(1.61)	(1.33)	(3.28)	(-3.35)
1963/07 – 1981/12	0.00	-0.05	0.05	0.18	-0.16	0.34	0.03	0.14	-0.11
	(0.00)	(-0.39)	(0.59)	(1.73)	(-0.66)	(1.75)	(1.41)	(2.09)	(-1.97)
1982/01 – 2006/12	0.01	-0.05	0.06	0.47	0.23	0.24	0.00	0.20	-0.19
	(0.16)	(-0.49)	(0.83)	(6.08)	(0.71)	(0.90)	(0.35)	(2.58)	(-2.76)

Table 6: Average Slopes of Industry Asymmetric Effects

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. In each June of year t, we formed BM for July of year t to June of year t+1 as the book value of the prior fiscal year divided by the market equity of the prior calendar year end. The book value is defined as stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. As in Fama and French (1992), the smallest and largest 0.5% of the BM observations are set to the next smallest or largest value of the variable. We use a firm's market equity in June of year t to measure its size for July of year t to June of year t+1. We exclude firms with negative book value and those that have not appeared on COMPUSTAT for at least two years. This table reports the time-series average of the coefficients of the following cross-sectional equation:

$$\begin{aligned} R_{it} &= \gamma_{0t} + \gamma_{1t}(MV_{it} - MV_{It}) + \gamma_{2t}(MV_{it} - MV_{It})I(MV_{it} > MV_{It}) \\ &+ \gamma_{3t}(BM_{it} - BM_{It}) + \gamma_{4t}(BM_{it} - BM_{It})I(BM_{it} > BM_{It}) \\ &+ \gamma_{5t}(P12M_{it} - P12M_{It}) + \gamma_{6t}(P12M_{it} - P12M_{It})I(P12M_{it} > P12M_{It}) + e_{it}, \end{aligned}$$

where MV_{It} refers to the median of market values of firms in industry I in month t; BM_{It} and $P12M_{It}$ are defined accordingly; and $I(A)$ denotes an indicator function, that takes the value of 1 when the statement A is true and 0 otherwise. The average slopes of the monthly cross-sectional regressions are reported; in the parenthesis, we report the Newey-West (1987) t-statistics adjusted for serial correlation and heteroscedasticity. Panels A, B, and C report the results of various subperiods for all months, January, and non-January, respectively.

	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	$\gamma_1 + \gamma_2$	$\gamma_3 + \gamma_4$	$\gamma_5 + \gamma_6$
Panel A: All months									
1963/07 – 2006/12	-0.26	0.25	0.27	0.07	-0.01	0.01	-0.01	0.34	0.00
	(-4.76)	(4.25)	(3.11)	(0.55)	(-0.41)	(0.22)	(-0.35)	(4.09)	(-0.14)
1963/07 – 1981/12	-0.18	0.09	0.07	0.33	-0.01	0.02	-0.09	0.40	0.01
	(-2.14)	(1.09)	(0.52)	(1.59)	(-0.17)	(0.25)	(-1.62)	(2.62)	(0.32)
1982/01 – 2006/12	-0.32	0.37	0.42	-0.12	-0.02	0.00	0.05	0.29	-0.01
	(-4.37)	(4.58)	(3.85)	(-0.77)	(-0.41)	(0.05)	(0.90)	(3.31)	(-0.69)
Panel B: January									
1963/07 – 2006/12	-1.78	0.85	-0.54	2.16	-0.92	0.99	-0.92	1.61	0.07
	(-5.90)	(3.05)	(-1.51)	(3.85)	(-5.80)	(4.69)	(-5.07)	(3.15)	(0.83)
1963/07 – 1981/12	-1.79	0.50	0.18	1.92	-0.81	0.77	-1.29	2.11	-0.05
	(-4.20)	(1.14)	(0.42)	(2.23)	(-2.89)	(2.05)	(-6.03)	(2.42)	(-0.27)
1982/01 – 2006/12	-1.77	1.11	-1.07	2.32	-0.99	1.14	-0.66	1.25	0.15
	(-4.39)	(3.37)	(-2.22)	(3.17)	(-5.50)	(4.64)	(-2.68)	(2.21)	(1.78)
Panel C: Non-January									
1963/07 – 2006/12	-0.13	0.20	0.34	-0.12	0.07	-0.08	0.07	0.23	-0.01
	(-2.35)	(3.33)	(3.78)	(-0.91)	(2.07)	(-1.53)	(1.81)	(3.07)	(-0.43)
1963/07 – 1981/12	-0.04	0.05	0.06	0.19	0.06	-0.04	0.01	0.25	0.02
	(-0.52)	(0.65)	(0.42)	(0.88)	(1.13)	(-0.51)	(0.19)	(1.82)	(0.45)
1982/01 – 2006/12	-0.19	0.30	0.55	-0.34	0.07	-0.10	0.11	0.21	-0.03
	(-2.69)	(3.73)	(4.91)	(-2.29)	(1.75)	(-1.67)	(2.19)	(2.62)	(-1.25)

Table 7: Average Slopes of Within- and Across-Industry Components and Industry Asymmetric Effects Using the Five-Factor Models as Adjustments

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. In each June of year *t*, we formed BM for July of year *t* to June of year *t*+1 as the book value of the prior fiscal year divided by the market equity of the prior calendar year end. The book value is defined as stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. As in Fama and French (1992), the smallest and largest 0.5% of the BM observations are set to the next smallest or largest value of the variable. We use a firm's market equity in June of year *t* to measure its size for July of year *t* to June of year *t*+1. We exclude firms with negative book value and those that have not appeared on COMPUSTAT for at least two years. This table reports the time-series average of the coefficients of the following cross-sectional equation:

$$\begin{aligned}
\tilde{R}^*_{it} &= \gamma_{0t} + \gamma_{1t}(MV_{it} - MV_{It}) + \gamma_{2t}(MV_{It} - \overline{MV_t}) + \gamma_{3t}(MV_{it} - MV_{It})I(MV_{it} > MV_{It}) \\
&+ \gamma_{4t}(BM_{it} - BM_{It}) + \gamma_{5t}(BM_{It} - \overline{BM_t}) + \gamma_{6t}(BM_{it} - BM_{It})I(BM_{it} > BM_{It}) \\
&+ \gamma_{7t}(P12M_{it} - P12M_{It}) + \gamma_{8t}(P12M_{It} - \overline{P12M_t}) \\
&+ \gamma_{9t}(P12M_{it} - P12M_{It})I(P12M_{it} > P12M_{It}) + e_{it},
\end{aligned}$$

where $\tilde{R}^*_{jt}$ is the risk-adjusted return; MV_{It} refers to the median of market values of firms in industry I in month t; $\overline{MV_t}$ refers to the median of MV_{It} in month t; BM_{It}, $P12M_{It}$, $\overline{BM_t}$ and $\overline{P12M_t}$ are defined accordingly; and $I(A)$ denotes an indicator function, which takes the value of 1 when the statement A is true and 0 otherwise. The average slopes of the monthly cross-sectional regressions are reported; in the parenthesis, we provide the Newey-West (1987) *t*-statistics adjusted for serial correlation and heteroscedastic. Panels A and B report the results of returns on individual stocks adjusted according to the industry-based five-factor model and the industry-augmented five-factor model, respectively.

		Size			BM			Momentum		
	γ_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_7	γ_8	γ_9
Panel A: Industry-based five-factor model										
All months	0.25	-0.48	-0.26	0.45	0.20	-0.18	-0.08	0.02	0.04	-0.09
	(2.54)	(-6.99)	(-4.10)	(6.31)	(2.43)	(-1.03)	(-0.69)	(0.77)	(0.92)	(-2.51)
January	0.79	-2.22	-1.25	1.57	-0.34	-0.34	0.74	-0.64	-0.57	0.45
	(1.94)	(-7.12)	(-4.84)	(4.88)	(-1.19)	(-0.69)	(1.58)	(-4.77)	(-4.15)	(3.32)
Non-January	0.20	-0.32	-0.17	0.34	0.25	-0.17	-0.15	0.08	0.10	-0.13
	(1.93)	(-4.86)	(-2.78)	(4.89)	(2.78)	(-0.90)	(-1.26)	(2.90)	(2.13)	(-4.09)
Panel B: Industry-augmented five-factor model										
All months	0.08	-0.36	-0.23	0.37	0.11	-0.36	-0.08	0.03	0.02	-0.09
	(1.08)	(-5.62)	(-3.94)	(5.45)	(1.40)	(-2.44)	(-0.72)	(0.93)	(0.42)	(-2.38)
January	0.28	-1.73	-1.27	1.27	-0.59	-1.52	0.48	-0.72	-0.72	0.42
	(1.11)	(-5.88)	(-4.34)	(4.53)	(-2.24)	(-3.20)	(0.99)	(-4.71)	(-3.64)	(2.71)
Non-January	0.06	-0.24	-0.13	0.28	0.18	-0.25	-0.13	0.10	0.09	-0.14
	(0.82)	(-3.85)	(-2.50)	(4.30)	(2.09)	(-1.72)	(-1.11)	(3.64)	(1.80)	(-3.88)

Table 8: Average Slopes of Within- and Across-Industry Components and Industry Asymmetric Effects with Industry Concentration

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. In each June of year *t*, we formed BM for July of year *t* to June of year *t*+1 as the book value of the prior fiscal year divided by the market equity of the prior calendar year end. The book value is defined stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. As in Fama and French (1992), the smallest and largest 0.5% of the BM observations are set to the next smallest or largest value of the variable. We use a firm's market equity in June of year *t* to measure its size for July of year *t* to June of year *t*+1. We exclude firms with negative book value and those that have not appeared on COMPUSTAT for at least two years. This table reports the time-series average of the coefficients of the following cross-sectional equation:

$$\begin{aligned}
\tilde{R}^*_{it} &= \gamma_{0t} + \gamma_{1t} H_{It}(S) + \gamma_{2t}(MV_{it} - MV_{It}) + \gamma_{3t}(MV_{It} - \overline{MV_t}) + \gamma_{4t}(MV_{it} - MV_{It}) I(MV_{it} > MV_{It}) \\
&+ \gamma_{5t}(BM_{it} - BM_{It}) + \gamma_{6t}(BM_{It} - \overline{BM_t}) + \gamma_{7t}(BM_{it} - BM_{It}) I(BM_{it} > BM_{It}) \\
&+ \gamma_{8t}(P12M_{it} - P12M_{It}) + \gamma_{9t}(P12M_{It} - \overline{P12M_t}) \\
&+ \gamma_{10t}(P12M_{it} - P12M_{It}) I(P12M_{it} > P12M_{It}) + e_{it},
\end{aligned}$$

where $H(S)$ refers to the Herfindahl index; MV_{It} is the median of market values of firms in industry I in month t; $\overline{MV_t}$ refers to the median of MV_{It} in month t; BM_{It}, $P12M_{It}$, $\overline{BM_t}$ and $\overline{P12M_t}$ are defined accordingly; and $I(A)$ denotes an indicator function, which takes the value of 1 when the statement A is true and 0 otherwise. The average slopes of the monthly cross-sectional regressions are reported; in the parenthesis, we provide the Newey-West (1987) *t*-statistics adjusted for serial correlation and heteroscedasticity. Panels A, B, and C report the results of unadjusted returns, returns on individual stocks adjusted based on the industry-based five-factor model, and returns adjusted according to the industry-augmented five-factor model, respectively.

			Size			BM			Momentum		
	γ_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_7	γ_8	γ_9	γ_{10}
Panel A: Raw return											
All months	1.38	-0.34	-0.33	-0.22	0.32	0.32	-0.12	-0.01	0.02	0.13	-0.03
	(5.17)	(-2.83)	(-4.94)	(-2.88)	(5.00)	(4.34)	(-0.58)	(-0.10)	(0.65)	(2.60)	(-0.74)
January	4.85	-1.05	-2.20	-1.81	1.22	-0.20	-0.98	1.21	-0.74	-0.55	0.70
	(4.04)	(-2.85)	(-6.28)	(-5.30)	(4.07)	(-0.58)	(-1.13)	(2.26)	(-5.35)	(-2.87)	(4.05)
Non-January	1.07	-0.27	-0.16	-0.08	0.24	0.37	-0.04	-0.12	0.09	0.19	-0.09
	(4.04)	(-2.25)	(-2.49)	(-1.09)	(3.75)	(4.76)	(-0.19)	(-1.30)	(3.56)	(3.78)	(-2.77)
Panel B: Industry-based five-factor model											
All months	0.30	-0.22	-0.49	-0.25	0.45	0.20	-0.17	-0.08	0.02	0.04	-0.09
	(2.79)	(-1.69)	(-6.94)	(-3.89)	(6.33)	(2.43)	(-1.00)	(-0.67)	(0.76)	(0.87)	(-2.45)
January	1.13	-0.77	-2.29	-1.32	1.63	-0.26	-0.31	0.65	-0.65	-0.60	0.46
	(2.67)	(-1.57)	(-7.44)	(-4.86)	(5.19)	(-0.83)	(-0.63)	(1.38)	(-5.04)	(-4.30)	(3.55)
Non-January	0.23	-0.17	-0.32	-0.15	0.35	0.25	-0.16	-0.14	0.08	0.10	-0.14
	(2.00)	(-1.34)	(-4.81)	(-2.46)	(4.88)	(2.71)	(-0.87)	(-1.18)	(2.91)	(2.16)	(-4.01)
Panel C: Industry-augmented five-factor model											
All months	0.18	-0.36	-0.38	-0.22	0.38	0.13	-0.34	-0.09	0.02	0.02	-0.08
	(2.23)	(-2.65)	(-5.70)	(-3.64)	(5.60)	(1.52)	(-2.39)	(-0.78)	(0.80)	(0.35)	(-2.17)
January	0.76	-1.17	-1.83	-1.34	1.36	-0.51	-1.44	0.38	-0.72	-0.75	0.44
	(2.14)	(-1.89)	(-6.41)	(-4.36)	(5.15)	(-1.73)	(-3.03)	(0.79)	(-4.89)	(-3.77)	(2.93)
Non-January	0.13	-0.29	-0.25	-0.11	0.29	0.19	-0.24	-0.13	0.09	0.09	-0.13
	(1.64)	(-2.29)	(-3.89)	(-2.13)	(4.36)	(2.14)	(-1.70)	(-1.09)	(3.47)	(1.83)	(-3.64)

Table 9: Average Slopes of Within- and Across-Industry Components and Industry Asymmetric Effects in Good and Bad States

In each month from July 1963 to December 2006, the industries were formed on the basis of the four-digit SIC codes from COMPUSTAT. In each June of year *t*, we formed BM for July of year *t* to June of year *t*+1 as the book value of the prior fiscal year divided by the market equity of the prior calendar year end. The book value is defined as stockholder equity plus balance sheet deferred taxes and investment credit (if available), minus the book value of preferred stock. As in Fama and French (1992), the smallest and largest 0.5% of the BM observations are set to the next smallest or largest value of the variable. We use a firm's market equity in June of year *t* to measure its size for July of year *t* to June of year *t*+1. We exclude firms with negative book value and those that have not appeared on COMPUSTAT for at least two years. This table reports the time-series average of the coefficients of the following cross-sectional equation:

$$\begin{aligned}
\tilde{R}^*_{it} &= \gamma_{0t} + \gamma_{1t}(MV_{it} - MV_{It}) + \gamma_{2t}(MV_{It} - \overline{MV_t}) + \gamma_{3t}(MV_{it} - MV_{It})I(MV_{it} > MV_{It}) \\
&+ \gamma_{4t}(BM_{it} - BM_{It}) + \gamma_{5t}(BM_{It} - \overline{BM_t}) + \gamma_{6t}(BM_{it} - BM_{It})I(BM_{it} > BM_{It}) \\
&+ \gamma_{7t}(P12M_{it} - P12M_{It}) + \gamma_{8t}(P12M_{It} - \overline{P12M_t}) \\
&+ \gamma_{9t}(P12M_{it} - P12M_{It})I(P12M_{it} > P12M_{It}) + e_{it},
\end{aligned}$$

where $\tilde{R}^*_{jt}$ is the risk-adjusted return; MV_{It} refers to the median of market values of firms in industry I in month t; $\overline{MV_t}$ refers to the median of MV_{It} in month t; BM_{It}, $P12M_{It}$, $\overline{BM_t}$ and $\overline{P12M_t}$ are defined accordingly; and $I(A)$ denotes an indicator function, which takes the value of 1 when the statement A is true and 0 otherwise. The average slopes of the monthly cross-sectional regressions are reported; in the parenthesis, we report the Newey-West (1987) *t*-statistics adjusted for serial correlation and heteroscedasticity. Panels A, B and C report the results of unadjusted returns, returns on individual stocks adjusted based on the industry-based five-factor model, and returns adjusted according to the industry-augmented five-factor model, respectively.

		Size			BM			Momentum		
	γ_0	γ_1	γ_2	γ_3	γ_4	γ_5	γ_6	γ_7	γ_8	γ_9
Panel A: Raw return										
Good states	1.94	-0.42	-0.37	0.28	0.26	-0.18	0.11	0.02	0.09	-0.03
	(4.24)	(-2.86)	(-2.05)	(2.23)	(1.90)	(-0.42)	(0.67)	(0.31)	(1.16)	(-0.45)
Bad states	0.38	-0.05	-0.03	0.10	0.39	-0.37	-0.44	0.06	0.29	-0.09
	(0.67)	(-0.39)	(-0.24)	(0.78)	(2.69)	(-0.94)	(-2.57)	(1.05)	(2.33)	(-1.09)
Difference	1.56	-0.36	-0.34	0.18	-0.13	0.19	0.54	-0.05	-0.20	0.06
	(2.23)	(-1.86)	(-1.58)	(0.99)	(-0.64)	(0.33)	(2.17)	(-0.61)	(-1.44)	(0.59)
Panel B: Industry-based five-factor model										
Good states	0.34	-0.46	-0.26	0.43	0.36	-0.07	-0.09	0.00	-0.01	-0.07
	(1.75)	(-3.59)	(-2.20)	(3.39)	(2.48)	(-0.20)	(-0.45)	(0.04)	(-0.08)	(-1.26)
Bad states	0.46	-0.36	-0.39	0.20	0.02	-0.76	-0.13	0.00	0.05	-0.01
	(2.43)	(-2.38)	(-2.97)	(1.40)	(0.10)	(-1.95)	(-0.62)	(-0.01)	(0.63)	(-0.09)
Difference	-0.12	-0.10	0.13	0.23	0.34	0.70	0.04	0.00	-0.06	-0.06
	(-0.41)	(-0.51)	(0.73)	(1.13)	(1.59)	(1.39)	(2.17)	(0.13)	(-0.52)	(-0.66)
Panel C: Industry-augmented five-factor model										
Good states	0.07	-0.42	-0.28	0.42	0.31	-0.02	-0.12	0.03	0.00	-0.09
	(0.46)	(-3.48)	(-2.59)	(3.22)	(2.39)	(-0.08)	(-0.59)	(0.59)	(-0.02)	(-1.61)
Bad states	0.24	-0.18	-0.15	0.13	-0.04	-0.91	-0.20	0.00	0.09	-0.06
	(1.72)	(-1.35)	(-1.23)	(0.97)	(-0.25)	(-2.90)	(-0.94)	(0.06)	(0.90)	(-0.81)
Difference	-0.17	-0.24	-0.12	0.30	0.35	0.89	0.08	0.03	-0.09	-0.03
	(-0.81)	(-1.35)	(-0.77)	(1.67)	(1.71)	(2.39)	(0.25)	(0.32)	(-0.70)	(-0.37)

Table A: Marginal and Cumulative Explaining Proportion of the Five Factors and Regressions of the Five Factors on the Three-Factor Model

The five factors are estimated as "unconditional" estimations from the 48 industry portfolios using principal components analysis. Marginal is the proportion of industry portfolio variation explained by each extracted factor. Cumulative is the sequential cumulative proportion of industry portfolio variation explained by the extracted factors 1 to 5. Regressions of the five extracted factors on the three- and four-factor models, respectively, are as follows:

$$F_i = a_i + b_i(R_m - R_f) + s_i SMB + h_i HML + e_i,$$

and

$$F_i = a_i + b_i(R_m - R_f) + s_i SMB + h_i HML + m_i MOM + \varepsilon_i,$$

for $i = 1, \ldots, 5$. The coefficients and t-statistics of regressions are reported. $Adj. R^2$ represents the adjusted R-square of regressions.

Factor	Marginal	Cumulative	a	b	s	h	m	$t(a)$	$t(b)$	$t(s)$	$t(h)$	$t(m)$	$Adj. R^2$
F1	0.58	0.58	0.00	0.03	0.01	0.01		-1.41	71.27	13.59	12.45		0.93
			0.00	0.03	0.01	0.01	0.00	0.15	72.94	14.29	11.82	-6.81	0.93
F2	0.05	0.63	-0.15	-0.02	0.24	0.35		-0.67	-0.45	3.42	4.29		0.04
			-0.27	-0.01	0.24	0.38	0.13	-1.22	-0.15	3.39	4.62	2.43	0.05
F3	0.04	0.68	-0.09	0.05	-0.05	0.45		-0.95	2.02	-1.64	12.30		0.26
			-0.17	0.06	-0.05	0.46	0.08	-1.69	2.42	-1.71	12.76	3.25	0.27
F4	0.03	0.70	0.20	0.06	-0.21	-0.08		5.18	6.28	-16.73	-5.38		0.36
			0.20	0.06	-0.21	-0.08	0.00	5.10	6.19	-16.71	-5.35	-0.28	0.36
F5	0.03	0.73	0.44	-0.01	-0.20	-0.49		2.24	-0.12	-3.23	-6.74		0.09
			0.29	0.01	-0.21	-0.46	0.16	1.44	0.28	-3.32	-6.24	3.26	0.10

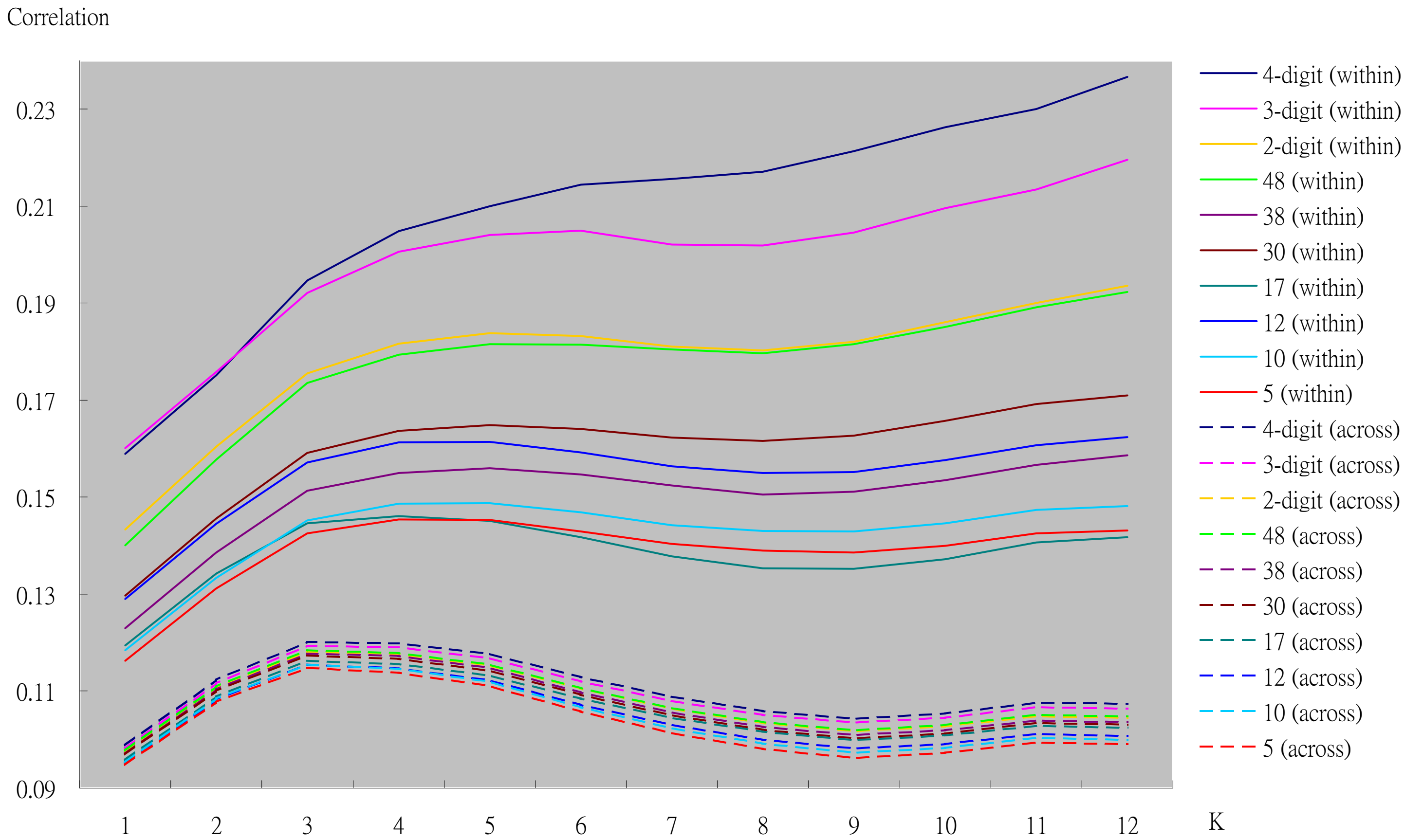

Figure 1: Within- and Across-Industry Correlations over Various Return Intervals